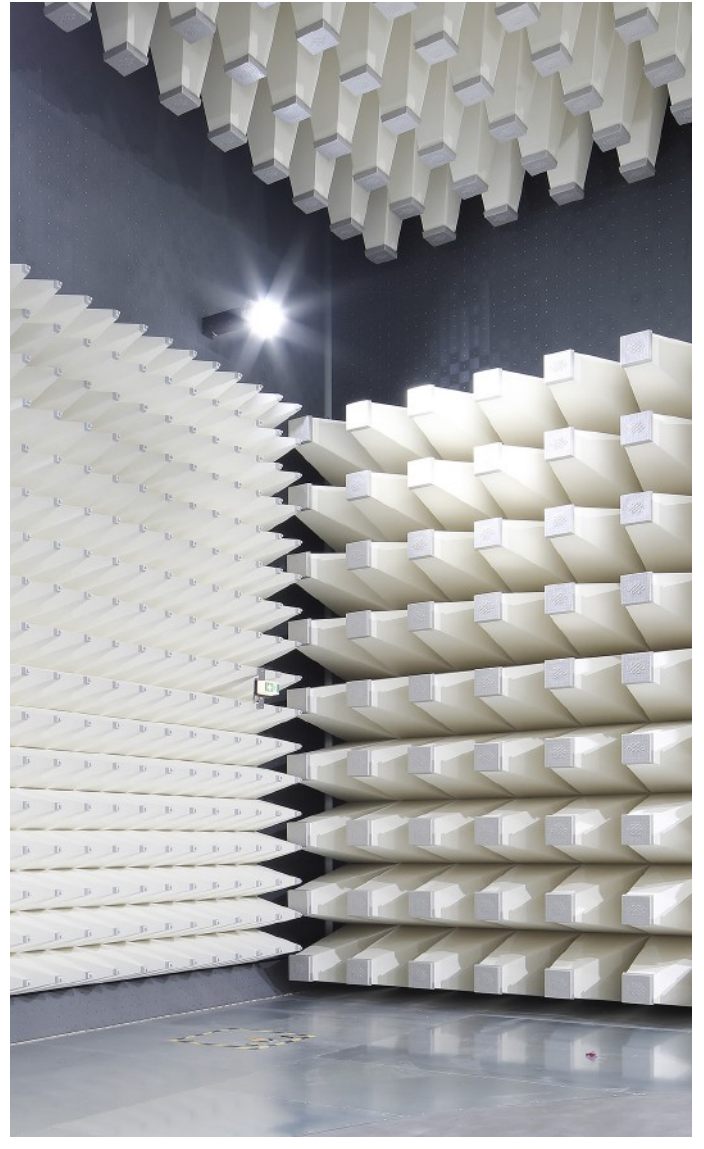
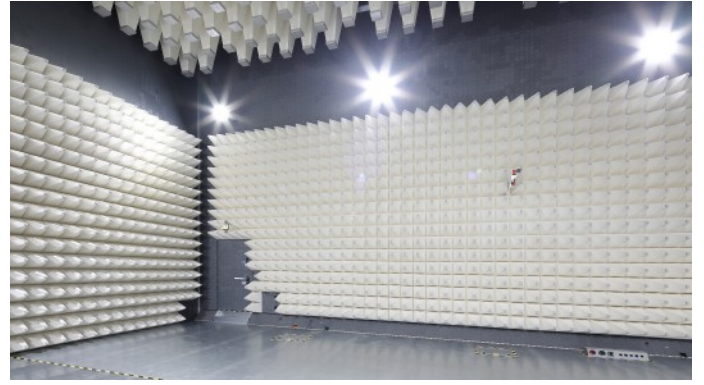
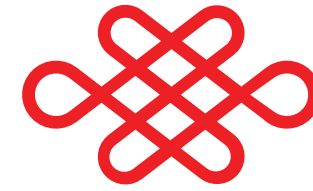




FRANKONIA
GROUP

ANECHOIC CHAMBERS

Customized Project & Chamber Solutions



FRANKONIA

GROUP

The Frankonia Group was founded in 1987 as a solution provider for EMC and antenna test laboratories, meeting the increasing demand for highly specialized testing environments for the electronic and automotive industry.

Without limitations in its capabilities, Frankonia develops future-oriented concepts for our complete product range, which guarantee the optimal use of resources, as well as the best possible customized solutions.

Frankonia offers complete solutions for the electronic, military and automotive industry, which meet customers' individual requirements.

Frankonia **stands for**

- › Pre-compliance and full compliance anechoic chamber solutions
- › Semi-anechoic and fully anechoic EMC Chambers
- › Standardized or custom-made chambers
- › Antenna Performance Chambers
- › Reverberation Chambers with own stirrer solutions
- › Individual EMC Test Systems
- › Innovative positioning and automation devices
- › Unique Frankosorb® absorber technology

Frankonia **trusted solutions**

- › Turnkey and future-proof solutions
- › Expertise in every stage of a project
- › Extended services: planning & consultation, testing readiness
- › Everything from one source
- › In-house engineering and manufacturing
- › High quality product scope
- › Global presence and worldwide activity

The EMC and antenna testing industry is a highly technical, innovative and fast-changing niche industry. With more than 35 years of experience to date, Frankonia maintains its leading position in laboratory solutions worldwide.

www.frankonia-solutions.com

Frankonia Group

The unique and trustworthy partner
for EMC and Antenna Solutions
worldwide.



Frankonia Group

Frankonia is recognized as a highly specialized technology corporation for EMC anechoic chambers and test systems within the automotive, military and industrial sectors for testing the electromagnetic compatibility.

Frankonia is the preferred supplier for complete solutions when it comes to the implementation of EMC test facilities. Passionate employees, many of them with longtime affiliation and experience, plan, coordinate, and define customized solutions with and for our customers that meet today's and future standards.

With our expertise, flexibility, quality and a high degree of technology, we generate future-proof solutions on a global scale. Due to our commitment to providing holistic EMC lab solutions, we offer outstanding expertise in every phase of a project.

Frankonia's project business convinces with its in-house project management, engineering and production, a trend-setting research, as well as its own installation and service teams. In this way, we ensure to provide a high level of technology and quality.

Table of Content

About Frankonia

› Solution Portfolio	Page 6
› Extended Services	Page 7
› Advantages & Benefits	Page 66

Commercial Chamber Solutions

› Shielded Room	Page 10
› CHC & CTC	Page 12
› FAC-3 & FAC-3 L	Page 16
› SAC-3 Plus	Page 18
› SAC-5 Plus	Page 20
› SAC-3 Square & SAC-5 Square	Page 24
› SAC-3/FAC-3 Transfomer	Page 26
› SAC-10 Plus Triton	Page 30
› SAC-10/H Hybrid	Page 34
› SAC-10/P Pyramid	Page 36

Automotive Chamber Solutions

› ACTC & UCC	Page 40
› AVTC	Page 42
› SAC-10V	Page 44
› EDTC & BlueBox	Page 48

Reverberation Chamber Solutions

› Reverberation Chambers	Page 50
--------------------------	---------

Military Chamber Solutions

› MIL CHC & MIL CPC	Page 54
› MIL-STD & MIL-STD Advanced	Page 54

Absorbers & Accessories

› Frankosorb® Absorbers	Page 58
› Shielding & Accessories	Page 62
› Automation	Page 64

Frankonia at a Glance

- › Frankonia demonstrates a global presence in cooperation, with a well-structured network of productions, representations and service units.
- › Frankonia provides fundamental knowledge to operate as a complete solution provider.
- › Frankonia implements innovative technologies to enhance efficiency and improve the outcomes and quality along with customers' needs.
- › Frankonia strives to be the preferred partner for customized and state-of-the-art solutions.

Frankonia's Anechoic Chambers are part of testing laboratories in different industries worldwide. Customers from commercial test institutes, manufacturers of electronic devices, as well as customers from the automotive or military industry trust Frankonia's solutions for now more than 35 years.

Solution Portfolio

We bring your project to perfection

Frankonia starts from the very first moment with planning, technical drawings, coordination and definition that meet our customers' individual demands.

Due to the solution ambitiousness, Frankonia provides excellent expertise in every stage of a project.

Project Planning

- › Frankonia provides precise knowledge of state-of-the-art Anechoic Chamber and Test System solutions with the latest technology and highest quality, while considering today's and future standards. Together with our customers, we define customized projects, provide technical details, project timelines and complete drawings. Herewith, in every single phase of the project, we ensure a perfect solution.

Project Management

- › From the first moment to the final handover, our project management takes care of the project and represents the central interface between Frankonia's scope of delivery and building related parties. Thus, we meet our customers' requirements without compromise.

Engineering

- › Frankonia's engineering ensures that our customers' requirements will be implemented impeccably from a single product up to a complete solution. Having our own engineering guarantees the quality and technology level that our customers expect. Herewith, Frankonia claims to be a competent partner.

Research & Development

- › Frankosorb® is Frankonia's unique absorber innovation that defines the leading standard for absorber technologies. Our continuous research on materials, a proper adaptation to our customers' requirements, as well as the respect of future standards put Frankonia's R&D department to the core when it comes to providing long-term efficient and trustful chamber solutions.

Manufacturing

- › Frankonia runs a stand-alone production network and constantly invests in innovative technologies and latest manufacturing equipment to meet our customers as well as our own expectations. Having our own manufacturing resources, which enable us to consistently provide the highest production quality and accuracy, distinguishes Frankonia from others. Moreover, it allows us to be independent and flexible in time.

Implementation

- › Our own European installation team ensures a proper implementation and installation of single products up to complete solutions that are engineered, developed and manufactured according to Frankonia's modular and prefabricated standards. Offering a complete scope of products and a broad range of international capabilities verifies Frankonia to be named as the trusted turnkey solution provider.

Solution

- › With more than 35 years of experience, Frankonia's goal is to transform our customers' individual requirements into reliable state-of-the-art solutions, always considering today's and future standards and test related specifications. We offer our expertise and innovation to customers seeking for complete solutions with leading ambitions. Because we know that only a complete and comprehensive solution can create and maintain long-term satisfaction.



Extended Services

We support your project with expertise

Planning & Consultation

For decades, Frankonia offers exclusive expertise to guide our customers through the entire course of the realization of a laboratory. Our non-stop solution offers consultancy from a first idea towards a complete testing readiness level. The benefit for our customers is that everything matches in terms of efficient, cost-saving and project management.

- › Layout of labs, practical usability, consultation on needs and requirements
- › Building planning in 3D for new facilities, or use of existing buildings
- › Infrastructure definition including planning of building services (ventilation, power, fire prevention, fire simulation)
- › Lab planning for/with architects, efficient implementation of test equipment parameters
- › Calculation on costs, return on investment, use and real-life practice to run a lab, profitability analysis

Testing Readiness

Together with our customers, we define the work packages, milestones for a modular, fast and reliable consulting and guidance to achieve your EMC goals. We offer a strong network of EMC experts and thus share standards and best practices from the international industrial, military or automotive world.

- › Test system and software trainings for rookies or professional, insights into EMC standards
- › Setup of hardware and software, functionality check, setup equipment for verification procedures and routines
- › Create appropriate test templates, test plans, implement measurement methods, or supervise your accreditation



Trusted

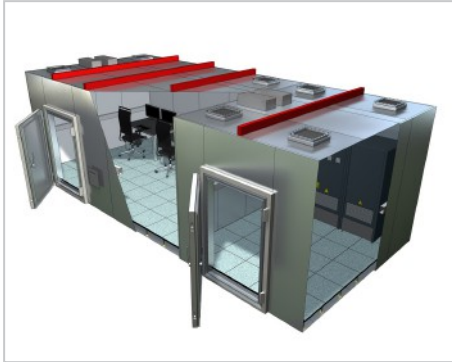
Shielded Room

Modular and Prefabricated PAN Type Shielding

Frankonia shielded rooms and anechoic chambers are designed based on a modular construction system. Prefabricated high quality shielding panels guarantee a maximum of flexibility regarding possible dimensions. All PAN type modules allow an easy handling and entry via standard building doors. The standard modules are bolted from inside every 75 mm with high conductivity mesh gasket inserted for sealing the joints of the panels. This facilitates an installation close to the walls of the parent building. The short screwing distance and the precise tightening of the screws with predefined torque guarantee long life shielding attenuation characteristics.

Features

- › PAN Type shielding modules made of 2,0 mm thick galvanized steel
- › Modular and prefabricated standard
- › Self-supporting stability or with static steel structure for any seismic condition
- › Mounted from the inside
- › Reverse installation possible (flat surface inside)
- › Interior finishing (walls and ceiling) possible
- › Raised floor systems, or welded floor systems
- › Long life shielding attenuation characteristics
- › No glue, no welding
- › Dismountable without any damage, easy modifications and maintenance
- › A complete transfer or future modification is possibility
- › Turnkey solution



Shielding standards

- › Frequency range acc. to EN 50147-1 or IEEE-299 (option) from 10 kHz up to 18 GHz or 40 GHz
- › Equal performance for any kind of feed-through components, honeycombs, doors and gates, filters, etc.
- › Perfectly adapted for Frankosorb® Absorbers
- › Any size of shielding is possible
- › Acoustic panels with absorption per ISO 354 $\alpha_w=0,65$ (MH)

Guaranteed Performance

Frequency	Attenuation	Field
10 kHz	≥ 90 dB	Magnetic Field
100 kHz	≥ 100 dB	Magnetic Field
1 MHz	≥ 110 dB	Magnetic Field
100 MHz	≥ 120 dB	Plane Wave
400 MHz	≥ 120 dB	Plane Wave
1 GHz	≥ 110 dB	Plane Wave
18 GHz	≥ 100 dB	Microwave
40 GHz	≥ 100 dB	Microwave

CHC & CTC

3m Compact Hybrid Chamber

The **CHC** is Frankonia's compact hybrid chamber solution at **3,0 m** measuring distance with a Quiet Zone (QZ) of **ø1,2 m**. It is an optimal solution for both pre-compliance emission tests and full compliant immunity tests at 3,0 m measuring distance. The extended version CHC L, includes an absorber-lined partition wall that offers the feature to house and store RF power amplifiers, antennas, or floor absorbers inside the chamber.

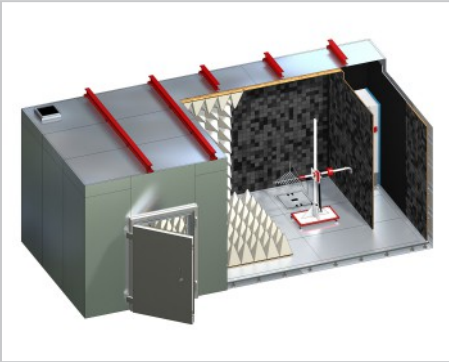
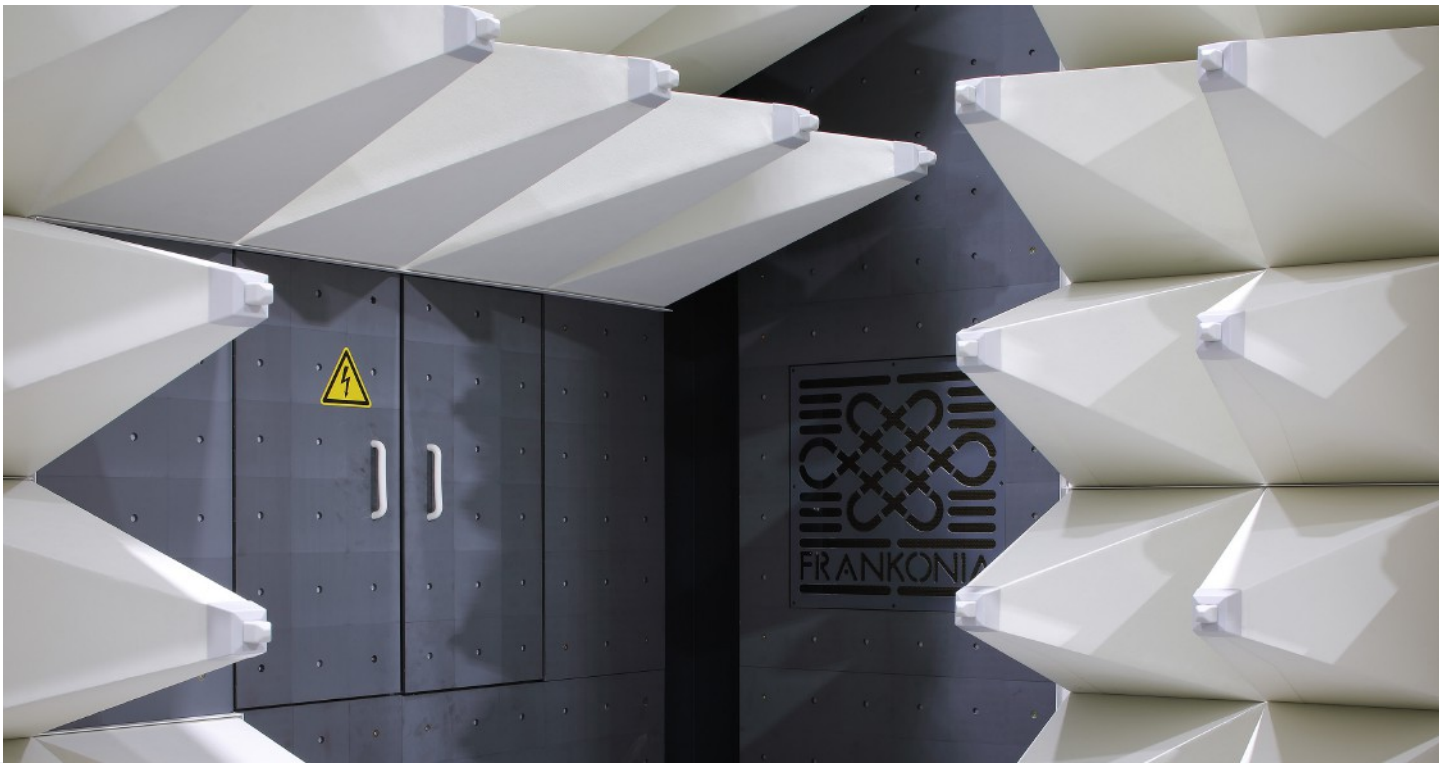
- › Pre-compliant EMI from 30 MHz to 1 GHz acc. to CISPR 16-1-4
- › Full compliant and cost saving solution for EMS acc. to IEC/EN 61000-4-3

The **CHC Plus** version is the advanced setup that allows compliant emission measurements from 1 GHz to 18 GHz.

- › Pre-compliant EMI from 30 MHz to 1 GHz acc. to CISPR 16-1-4 and compliant EMI from 1 GHz to 18/40 GHz
- › Full compliant and cost saving solution for EMS acc. to IEC/EN 61000-4-3

The **CTC** is Frankonia's full compliant component test chamber that entirely focus to immunity testing for industrial products, paired with automotive component EM and EMS, as well as military tests.

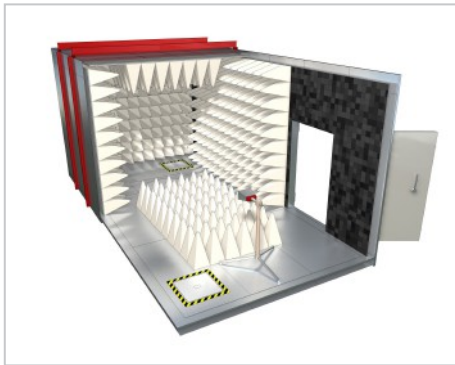
- › Full compliant and cost saving solution for EMS acc. to IEC/EN 61000-4-3
- › Full compliant with CISPR 25 and ISO 11452
- › Full compliant with MIL-STD 461 and DO-160



CHC
Pre-compliant EMI
Full compliant EMS



CHC Plus
Semi-compliant EMI
Full compliant EMS



CTC
Full compliant EMS
Full compliant CISPR 25
Full compliant MIL-STD 461

CHC	7,355 x 3,755 x 3,300 m (L x W x H) QZ ø1,2 m at 3,0 m test distance
CHC L	8,255 x 3,755 x 3,300 m (L x W x H) QZ ø1,2 m at 3,0 m test distance Feature: e.g., amplifier can be stored in the chamber
CHC Plus	7,355 x 3,755 x 3,300 m (L x W x H) QZ ø1,2 m at 3,0 m test distance Feature: Compliant emission >1 GHz
CHC Plus L	7,580 x 4,655 x 4,350 m (L x W x H) QZ ø1,2 m at 3,0 m test distance Feature: Turntable ø2,0, compliant emission >1 GHz
CTC	8,480 x 5,485 x 3,750 m (L x W x H) Full compliant immunity testing per IEC 61000-4-3 Feature: Full compliant to CISPR 25 and ISO 11452, MIL-STD 461 and DO-160
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Performance & Compliance

- › Pre-compliant emission (EMI) according to CISPR 16-1-4
Deviation NSA ±4,0 dB (30 MHz to 1 GHz) with limited hieght scan
- › Compliant emission (EMI) according to CISPR 16-1-4
Deviation SVSWR +6,0 dB (1 GHz to 18 GHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H450 or H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)



The image shows a large industrial facility, likely a testing or manufacturing environment. A prominent red overhead crane system is visible, spanning across the top of the frame. Below the crane, a large, light gray metal door is open, revealing a dark interior space with a grid-like structure, possibly an anechoic chamber. The floor is a smooth, light gray. To the left and right of the door, there are yellow safety barriers. The ceiling is high and features various pipes, ducts, and lighting fixtures. A red rectangular box with the word "Experienced" is overlaid on the lower left side of the image.

Experienced

FAC-3 & FAC-3 L

3m Fully Anechoic Chamber

The **FAC-3** is Frankonia's compact fully anechoic chamber at **3,0 m** measuring distance for EMC tests on **table-top positioned EUT's** with a Quiet Zone (QZ) of **ø1,5 m** (H= 1,5 m).

The **FAC-3 L** is the extended version of Frankonia's fully anechoic chamber at **3,0m** measuring distance with a Quiet Zone (QZ) of **ø1,5 m** (H= 2,0 m) for EMC tests on **table-top positioned** as well as on **floor-standing EUT's**.

Both are designed for measurements under free-space conditions based on CISPR 16-1-4 as a test site without ground plane. With its specific requirements for the test site, Frankonia's FAC-3 and FAC-3 L are supremely prepared to meet our customers' demands. The FAC-3 L offers in addition a height scan possibility using a FAM or FBM antenna mast.

Features

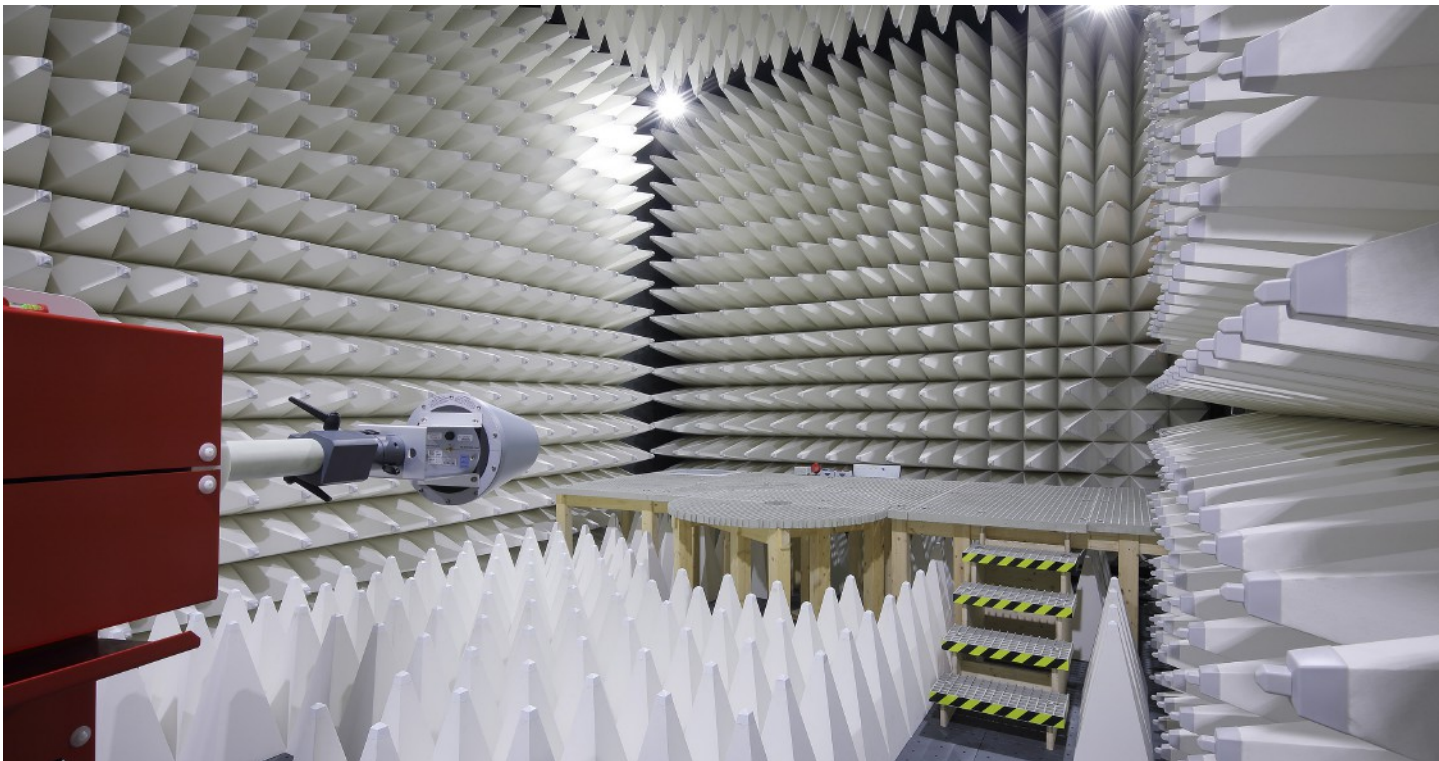
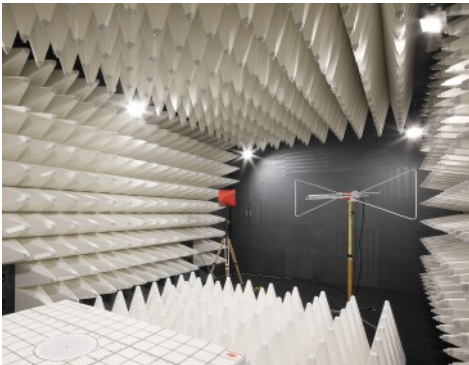
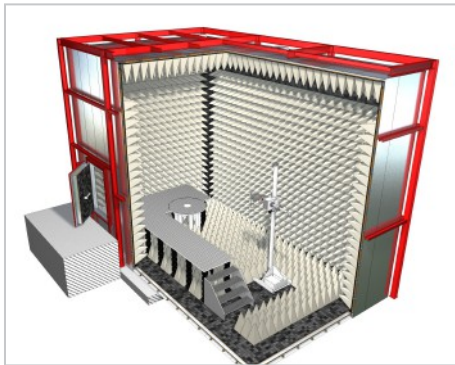
- › **FAC-3:** Test site for table-top EUT's
- › **FAC-3 L:** Test site for table-top and floor-standing EUT's (with height scan)
- › Full compliant EMI acc. to CISPR 16-1-4, IEC/EN 61000-4-22, and ETSI
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Cost-effective solution for free-space measurements
- › Compact chamber design with advanced Frankosorb® absorber lining
- › Double test axis option



FAC-3
Table-top EUT's



FAC-3 L
Floor-standing & table-top EUT's



FAC-3	8,705 x 4,655 x 3,750 m (L x W x H) QZ ø1,5 m at 3,0 m test distance (H= 1,5 m) Table-top products
FAC-3 L	9,380 x 5,780 x 6,000 m (L x W x H) QZ ø1,5 m at 3,0 m test distance (H= 2,0 m) Floor-standing & table-top products with height scan
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000 and H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4
Deviation FS NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)
- › Full compliant immunity (EMS) and emission (EMI) according to IEC/EN 61000-4-22
Deviation SdB c ≤ 1,8 dB

SAC-3 Plus

3m Semi Anechoic Chamber in Dome Design

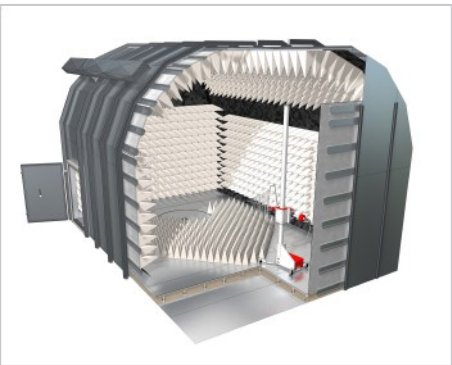
The **SAC-3 Plus** is Frankonia’s most versatile full compliant EMC testing solution at **3,0 m** measuring distance with a Quiet Zone (QZ) **up to ø2,0 m**. It is adapted for full compliant emission and immunity testing.

The innovatively shaped roof, called dome design, with its optimized Frankosorb® absorber layout leads to minimized reflections and offers outstanding performance for NSA, SVSWR and FU.

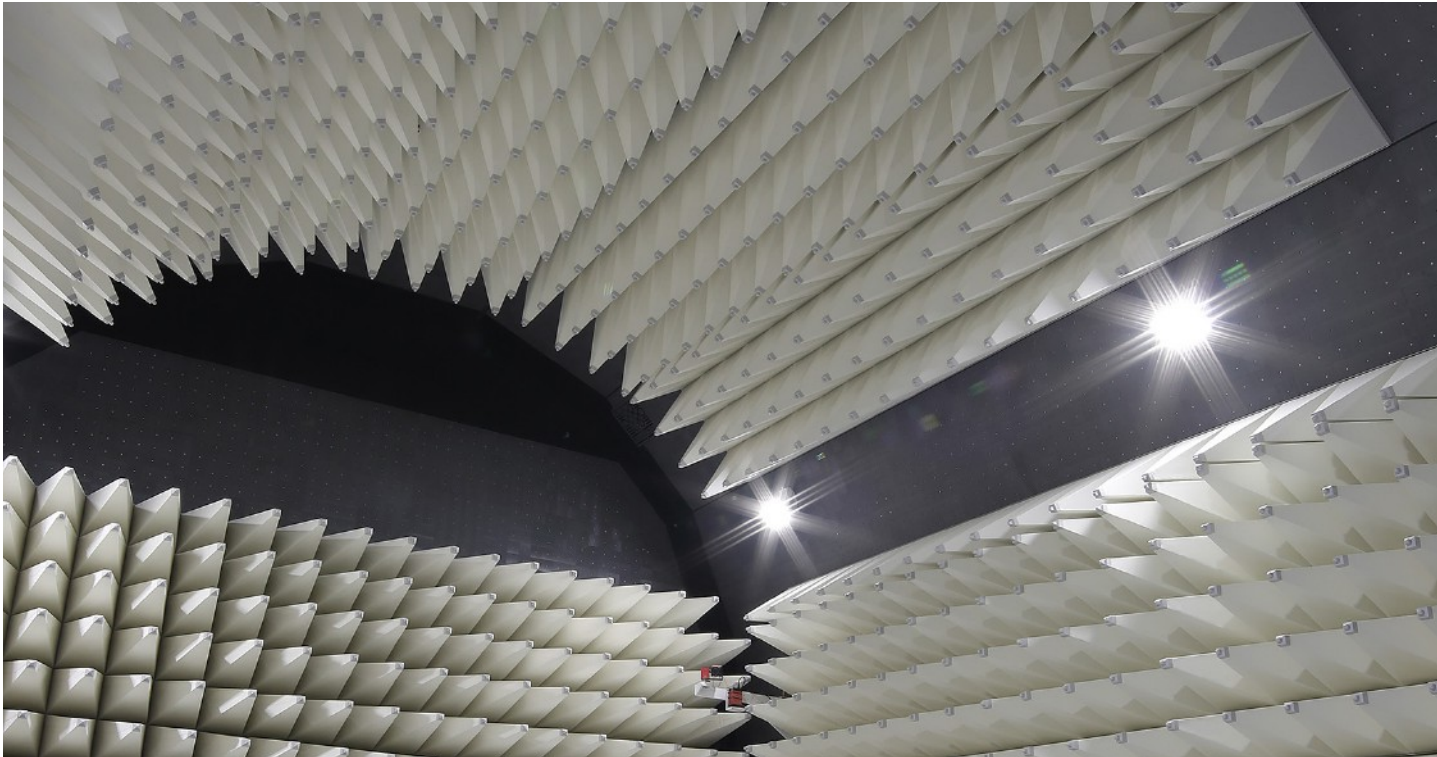
Features

- › Cost-effective high-performance solution for a 3,0 m measuring distance and QZ from ø1,2 m of up to ø2,0 m
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4 (ETSI upgradeable)
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Adapted lightweight steel structure and optimized RF-shielding
- › Innovative dome-shaped roof design
- › Upgradeable for E-Drive (load machine, BlueBox, battery test system)
- › Outstanding performance with long-lasting Frankosorb® hybrid absorbers
- › Usable for automotive and military standard tests

Since its introduction, the SAC-3 Plus has been the undisputed most selected chamber in its class. Through the innovative concept, customization and the excellent performance, it represents an efficient and economical solution that fully satisfies our customers.



SAC-3 Plus
Unique dome design



SAC-3 Plus S	8,480 x 6,530 x 6,000 m (L x W x H) QZ ø1,2 m at 3,0 m test distance (H= 2,0 m)
SAC-3 Plus M	8,780 x 6,530 x 6,000 m (L x W x H) QZ ø1,5 m at 3,0 m test distance (H= 2,0 m)
SAC-3 Plus L	9,230 x 6,530 x 6,000 m (L x W x H) QZ ø2,0 m at 3,0 m test distance (H= 2,0 m)
SAC-3 Plus	9,680 x 6,530 x 6,000 m (L x W x H) QZ ø2,0 m at 3,0 m test distance (H= 2,0 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000 and H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)

SAC-5 Plus

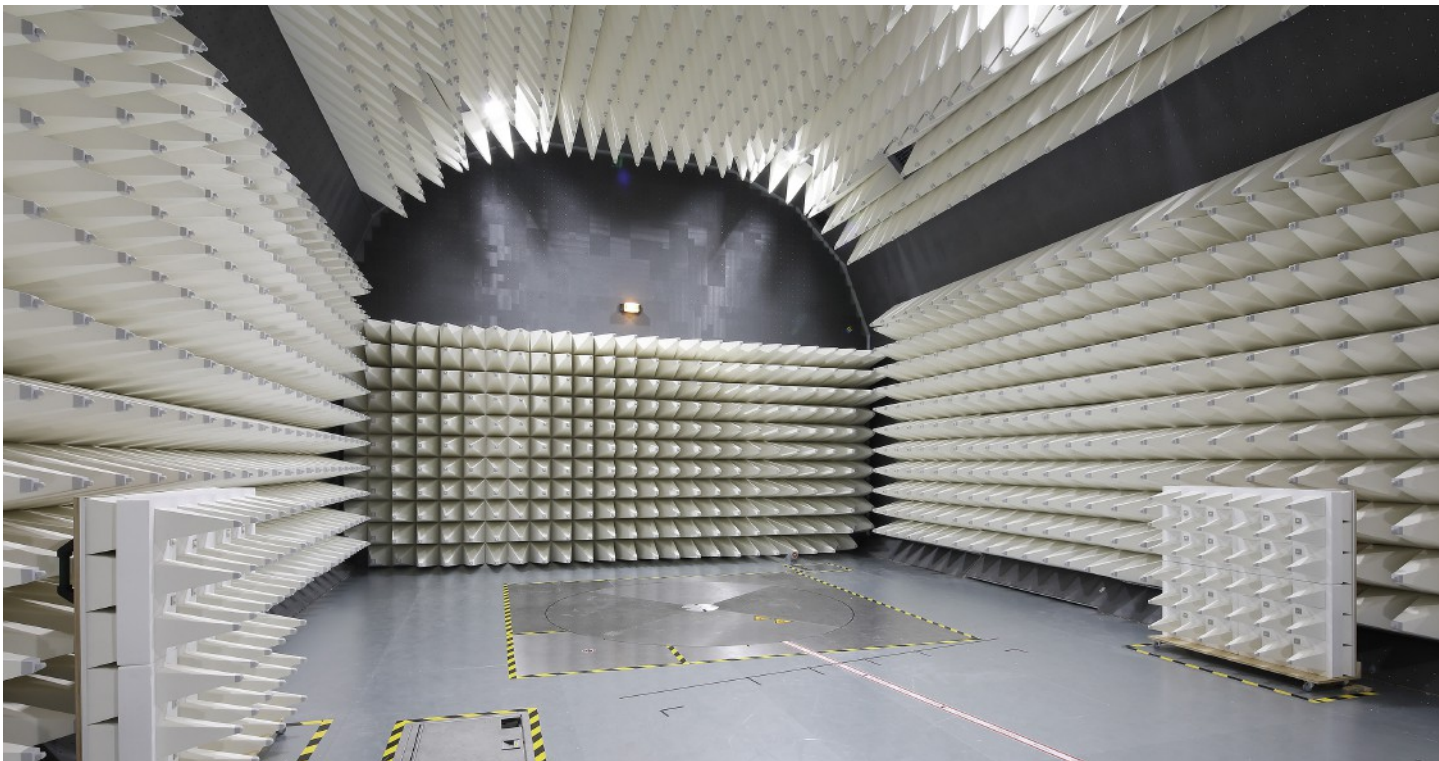
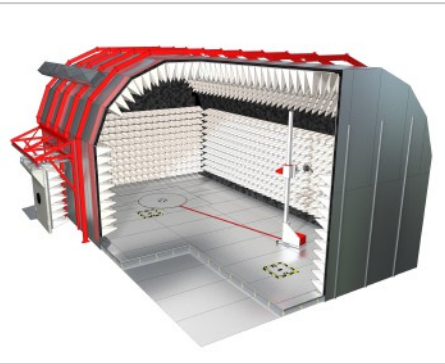
5m Semi Anechoic Chamber in Dome Design

The **SAC-5 Plus** is Frankonia’s full compliant EMC testing solution at **3,0 m** and **5,0 m** measuring distance with a Quiet Zone (QZ) **up to ø3,0 m**. It offers an innovative concept with its dome shaped roof, customization and performance, and therefore represents an efficient and economical solution that fully satisfies our customers.

Features

- › Efficient high-performance solution for a 3,0 m and 5,0 m measuring distance and QZ from ø2,0 m of up to ø3,0 m
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4 (ETSI upgradeable)
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Adapted lightweight steel structure and optimized RF-shielding
- › Innovative dome-shaped roof design
- › Upgradeable for E-Drive (load machine, BlueBox, battery test system)
- › Outstanding performance with long-lasting Frankosorb® absorbers
- › Usable for automotive and military standard tests
- › Turnkey solution
- › Double test axis option

Following the success of the SAC-3 Plus, the SAC-5 Plus with its 5,0 m test distance is based on the same innovative dome-shape concept. Herewith, it represents as well an efficient and economical solution that fully satisfies our customers needs.



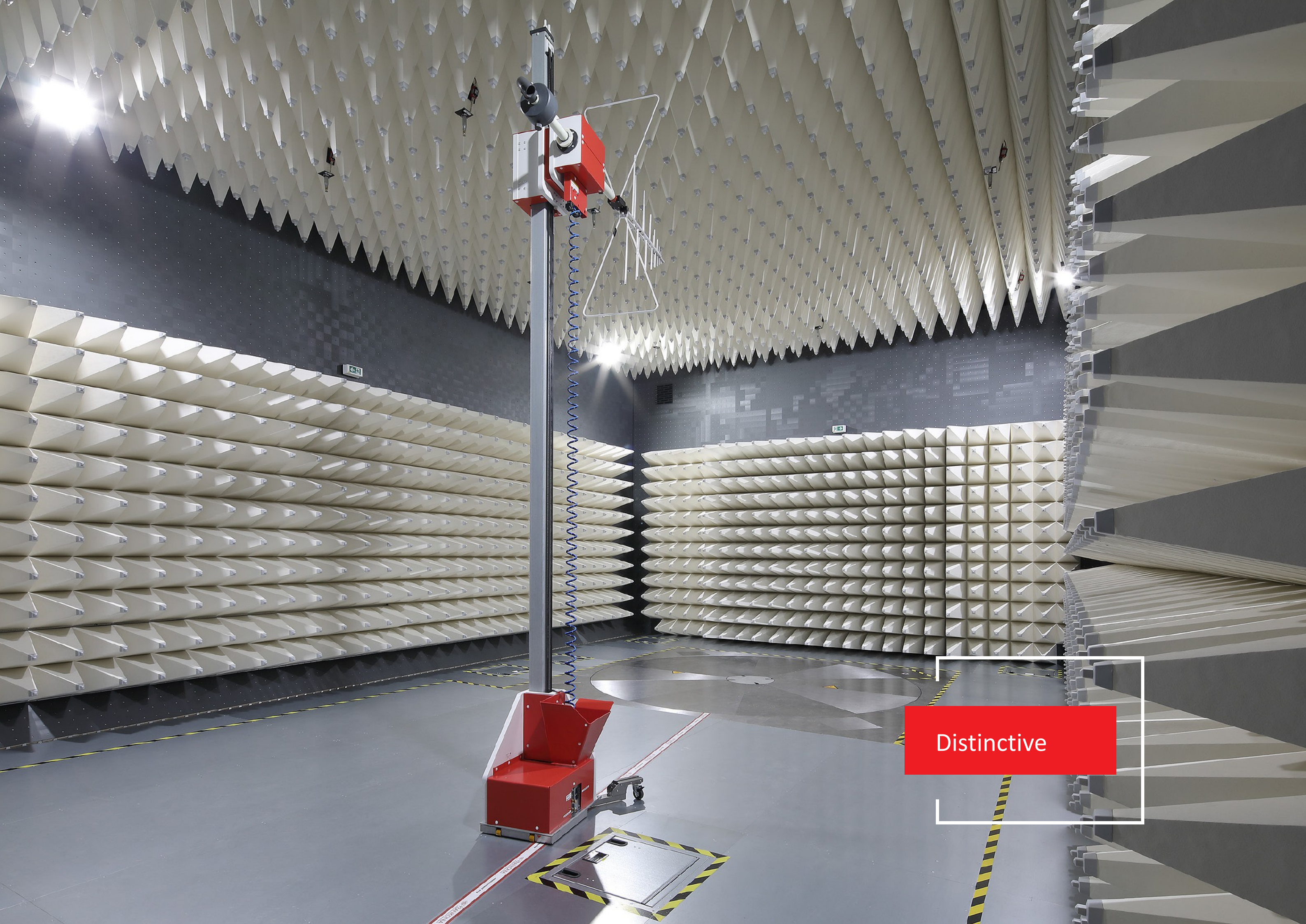
SAC-5 Plus	12,680 x 7,730 x 6,300 m (L x W x H) QZ ø2,0 m at 3,0 m & 5,0 m test distance (H= 2,5 m)
SAC-5 Plus L	12,680 x 8,180 x 6,300 m (L x W x H) QZ ø3,0 m at 3,0 m & 5,0 m test distance (H= 2,5 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000 and H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)



Distinctive

SAC-3 Square & SAC-5 Square

3m or 5m Semi Anechoic Chamber in Square Design

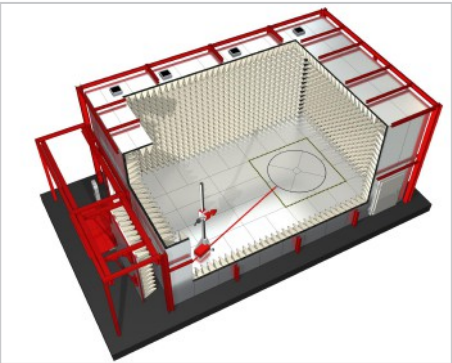
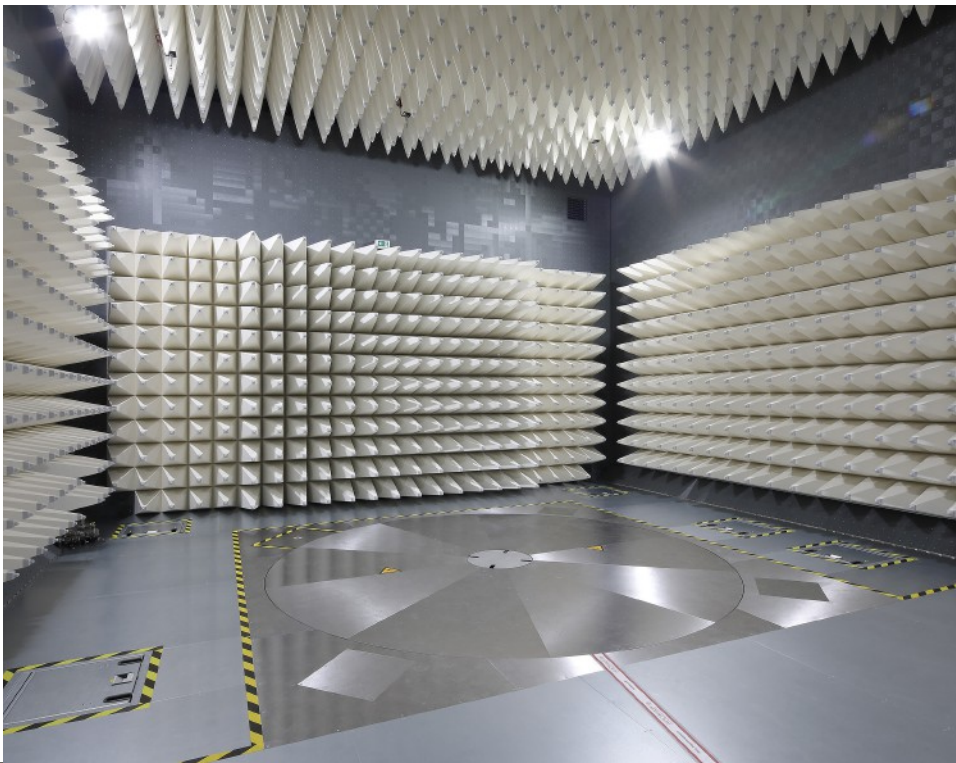
The **SAC-3 Square** is Frankonia’s versatile full compliant EMC testing solution at **3,0 m** measuring distance.

The **SAC-5 Square** offers a **3,0 m** and **5,0 m** measuring distance. Both models are available with a Quiet Zone (QZ) of **ø2,0 m** or **ø3,0 m**.

The SAC-3 and the SAC-5 in square design offer an innovative concept with its usability, customization and performance, and therefore represent efficient and economical solutions.

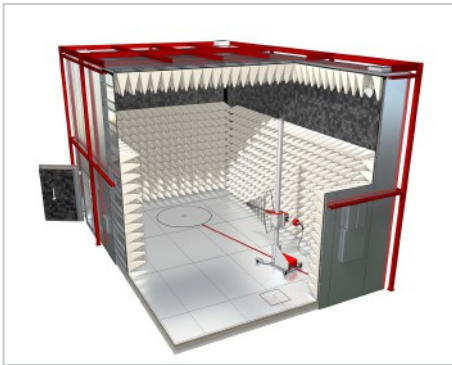
Features

- › Traditional square design
- › High-performance solution for 3,0 m or 5,0 m measuring distance
- › QZ from ø2,0 m up to ø4,0 m
- › Large turntable or mobile dynamometer integration
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4 (ETSI upgradeable)
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Immunity floor absorber storage in the chamber on trolley’s
- › Upgradeable for E-Drive (load machine, BlueBox, battery test system)
- › Outstanding performance with long-lasting Frankosorb® absorbers
- › Usable for automotive and military standard tests
- › Turnkey solution



SAC-5 Square
Traditional design

SAC-3 Square
Traditional design



Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H450 or H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

SAC-3 Square	9,680 x 6,530 x 6,000 m (L x W x H) QZ ø2,0 m at 3,0 m test distance (H= 2,5 m)
SAC-3 Square L	10,880 x 6,980 x 6,000 m (L x W x H) QZ ø3,0 m at 3,0 m test distance (H= 2,5 m)
SAC-5 Square	12,680 x 7,730 x 6,000 m (L x W x H) QZ ø2,0 m at 3,0 m & 5,0 m test distance (H= 2,5 m)
SAC-5 Square L	12,680 x 8,180 x 6,000 m (L x W x H) QZ ø3,0 m at 3,0 m & 5,0 m test distance (H= 2,5 m)
SAC-5 Square XL	13,280 x 9,380 x 6,300 m (L x W x H) QZ ø4,0 m at & 5,0 m test distance (H= 2,5 m) QZ ø3,0 m at & 3,0 m test distance (H= 2,5 m) Ready for a larger turntable or mobile dynamometer
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)

SAC-3/FAC-3 Transformer

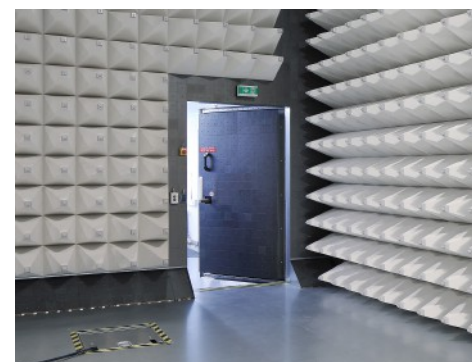
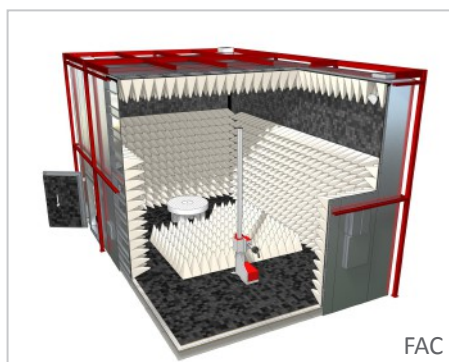
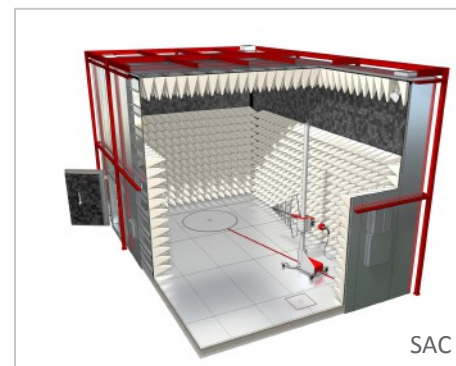
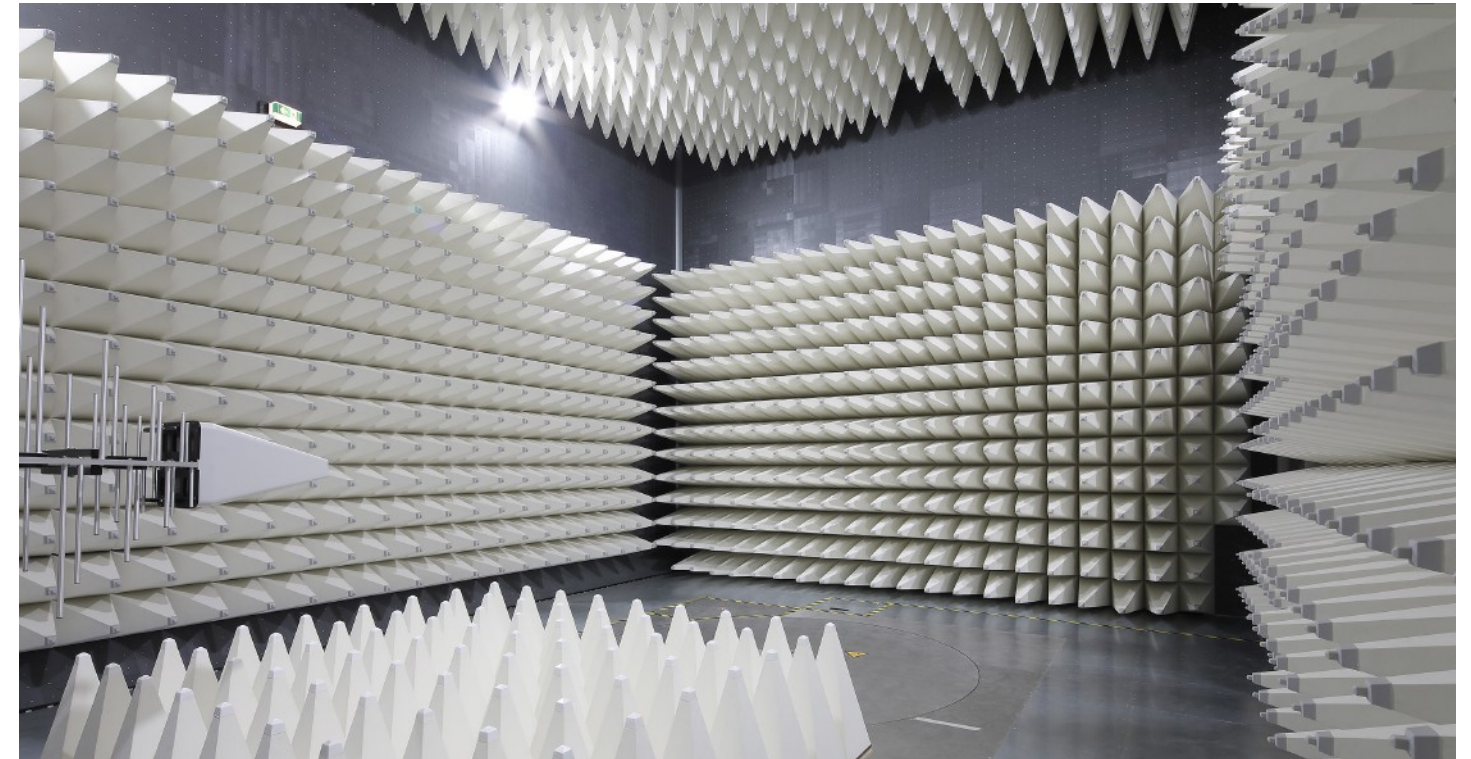
Semi & Fully Anechoic Chamber

Frankonia's **SAC-3/FAC-3 Transformer** is a full compliant EMC solution at **3,0 m** measuring distance offering semi as well as fully conditions.

This special solutions focus on conditions with ground plane, as well as FAR conditions for table-top EUT tests with an optimized floor absorber modification kit. The SAC-3/FAC-3 Transformer model is adapted to full compliant emission and immunity testing with a traditional square design.

Features

- › Cost-effective and high-performance solution for a 3,0 m test distance
- › SAC Setup: QZ of $\varnothing 2,0$ m for floor standing products
- › FAC Setup: QZ of $\varnothing 1,5$ m for table-top products
- › Full compliant EMI acc. to CISPR 16-1-4, ANSI C63.4, IEC/EN 61000-4-22, ETSI
- › Full compliant acc. to CISPR 25 and MIL-STD 461
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Upgradeable for E-Drive (load machine, BlueBox, battery test system)
- › Compact chamber design with advanced absorber lining
- › Outstanding performance with long-lasting Frankosorb® absorbers
- › Usable for automotive and military standard tests
- › Turnkey solution



SAC-3/FAC-3 Transformer	9,680 x 6,530 x 6,000 m (L x W x H) SAC: QZ $\varnothing 2,0$ m at 3,0 m test distance (H= 2,5 m) FAC: QZ $\varnothing 1,5$ m at 3,0 m test distance (H= 1,5 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

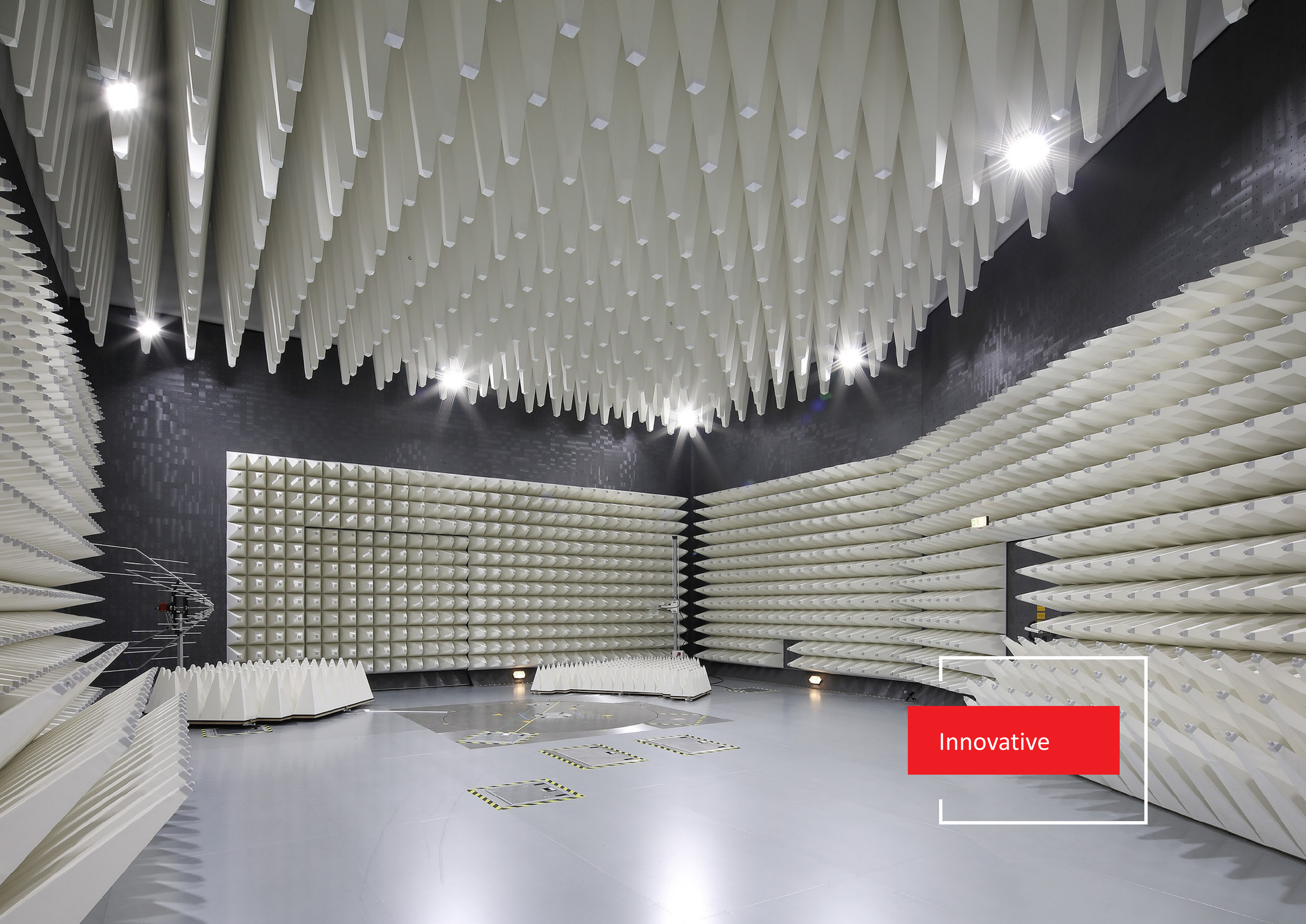
- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000 and H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance SAC - Semi Configuration

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA $\pm 3,5$ dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL $\pm 4,0$ dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0 dB/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)

Performance & Compliance FAC - Fully Configuration

- › Full compliant emission (EMI) according to CISPR 16-1-4
Deviation FS NSA $\pm 3,5$ dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0 dB/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)



Innovative

SAC-10 Plus Triton

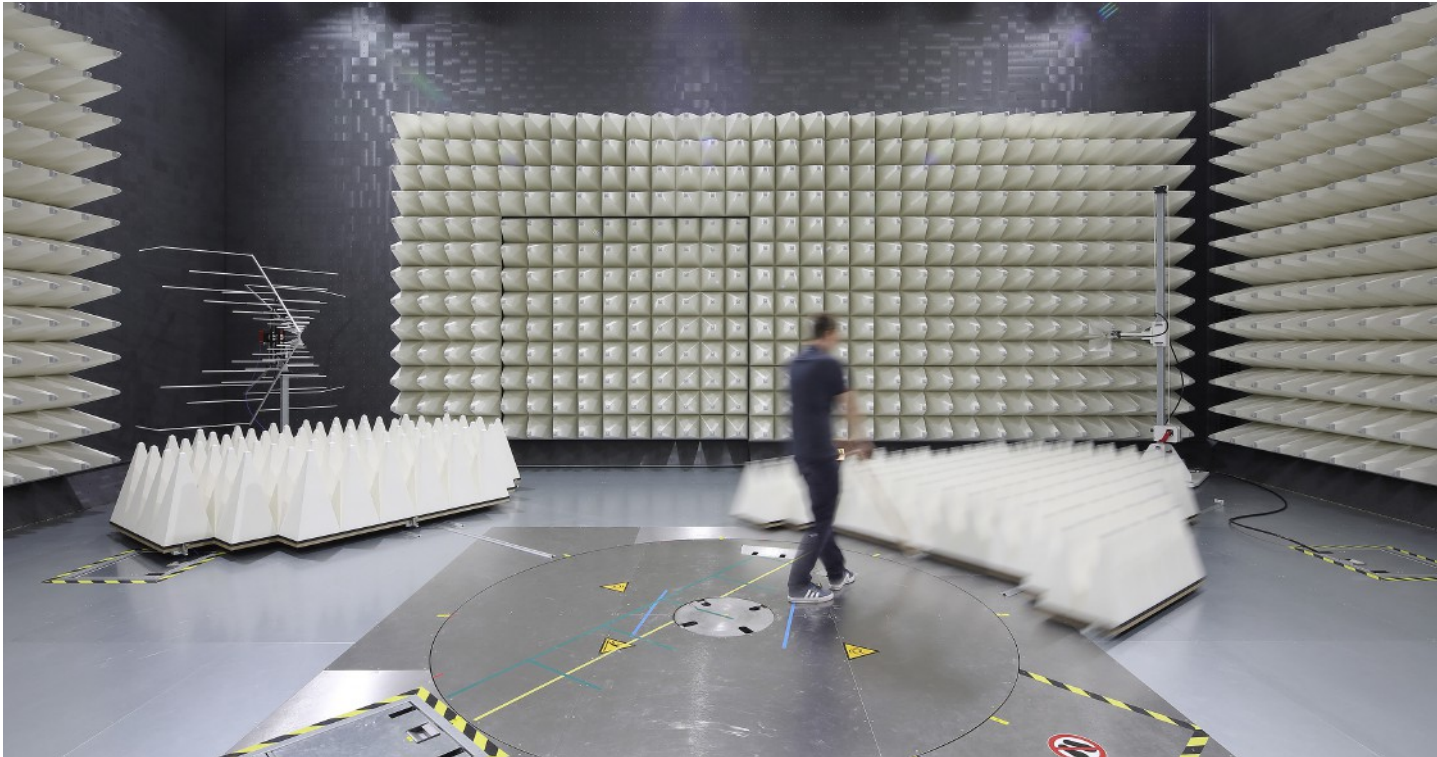
10m Advanced Semi Anechoic Chamber Solution

The **SAC-10 Plus Triton** is Frankonia’s full compliant state-of-the-art EMC testing solution with **multiple test axes** with **1x 10,0 m** and **2x 3,0 m** measuring distances with a Quiet Zone (QZ) of **ø3,0 m**. The innovative polygonal shape along with its optimized Frankosorb® absorber layout is a space-saving, cost-saving and efficient solution with a **multiple test axes**.

Features

- › Multiple test axes chamber with a Quiet Zone of ø3,0 m
- › 1x 10,0 m test distance and 2x 3,0m test distance in one solution
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4
- › Full compliant acc. to CISPR 25 and MIL-STD 461
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Space-saving and compact chamber design with polygonal shape
- › Floor absorbers and antennas remain connected in the chamber
- › Reproducibility and stable performance
- › Time-saving test setup with improved workflow and efficiency
- › Ingenious lining with long-lasting Frankosorb® non-combustible absorbers
- › Cost-saving and future-proof investment

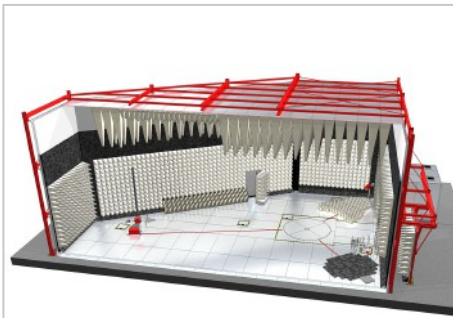
The SAC-10 Plus Triton is the most compact and lightweight 10 m chamber existing. It is full compliant for emission tests validated according to CISPR 16-1-4 and ANSI C63.4, as well as full compliant for immunity tests according to IEC/EN 61000-4-3, CISPR 25 and MIL-STD 461.



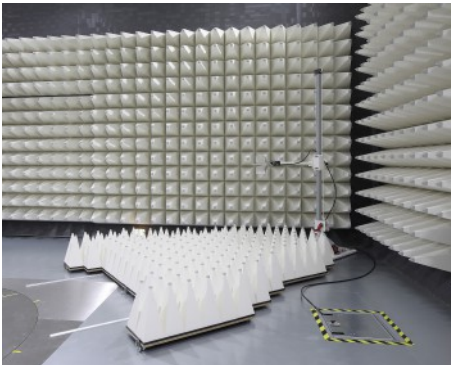
Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H450 or H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

SAC-10 Plus
Single test axis
cost-saver



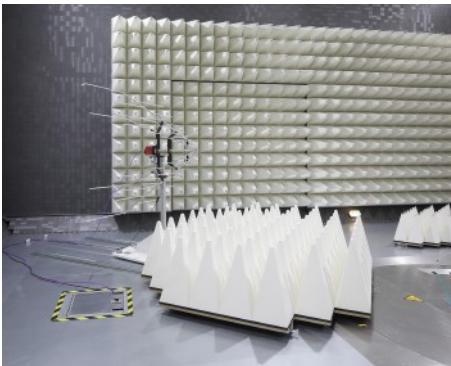
SAC-10 Plus Triton





SAC-10 Plus

SAC-10 Plus Triton
Multiple test axis
advanced solution



SAC-10 Plus Triton

- › Multiple test axes up to 10,0 m with a Quiet Zone of ø3,0 m
- › Single test axis up to 10,0 m with a Quiet Zone of ø3,0 m
- › With the use of three independent test axes dedicated to each EMI and EMS test, all antennas and floor absorbers remain connected in the chamber.
- › The Triton works like a regular SAC-10 chamber without any limitations.
- › Thanks to the guided floor absorber arrangement, the test setup preparation time reduces enormously, while reproducibility and quality of each testing remain stable.

SAC-10 Plus Triton	19,205 x 12,080 x 8,325 m (L x W x H) QZ ø3,0 m with multiple test axes (H= 3,0 m) 1x 10,0 m test distance (axis 1 = EMI and EMS) 1x 3,0 m test distance (axis 2 = EMI and EMS) 1x 3,0 m test distance (axis 3 = EMS)
SAC-10 Plus	19,205 x 12,080 x 8,325 m (L x W x H) QZ ø3,0 m at 10,0 m test distance (H= 3,0 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (30/80 MHz to 18 GHz)



Future-proof

SAC-10/H Hybrid

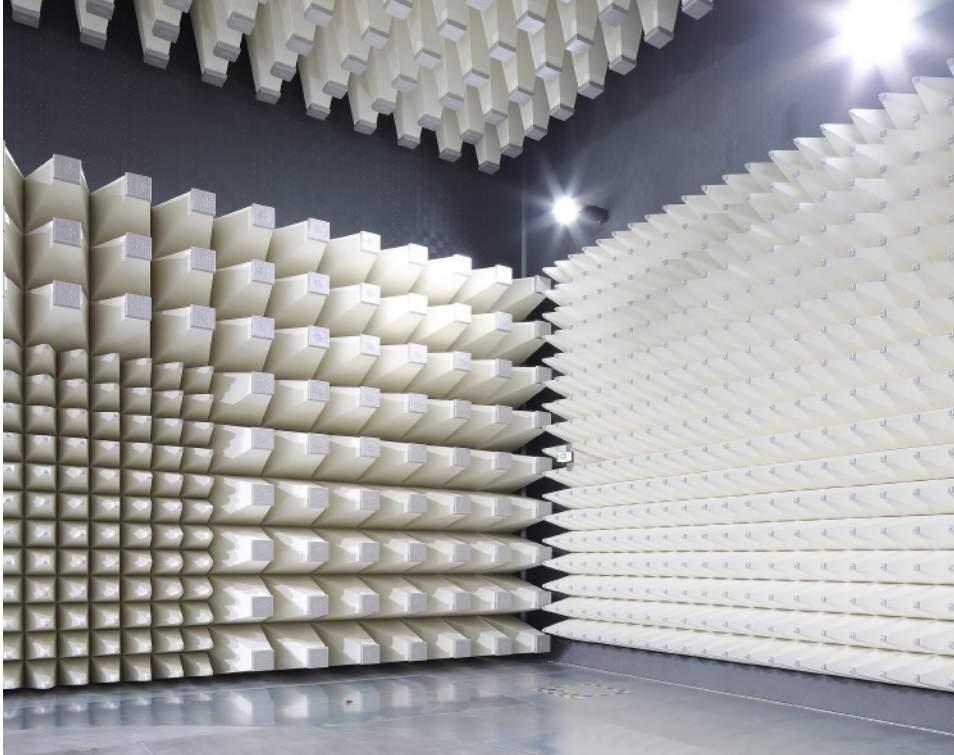
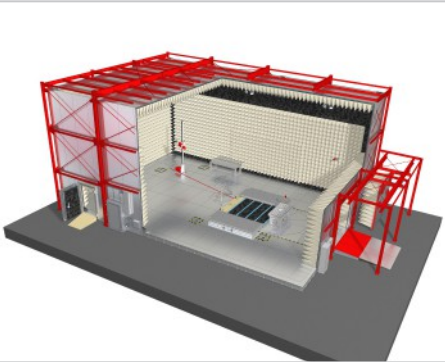
10m Semi Anechoic Chamber with Hybrid Absorbers

The **SAC-10/H** is Frankonia’s full compliant and customizable EMC testing solution at **10,0 m** measuring distance with a Quiet Zone (QZ) of **ø3,0 m** up to **ø6,0 m** and **hybrid** absorber layout.

Due to the high grade of customization reflecting the demands of our customers, this semi anechoic chamber is adaptable in size and offers several configuration possibilities. The impressionable Frankosorb® hybrid absorber layout achieves exceptional performance for emission measurements and immunity testing.

Features

- › Optimized lining with long-lasting and non-combustible Frankosorb® hybrid absorbers (Frankonia technology)
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Full compliant with military and automotive standards
- › Highly customizable solution for any kind of EMC testing and limitless integration of individual applications; as single or double test axis option
- › Notably adjustable anechoic chamber size, characteristics and configuration due to different EUT requirements
- › Advanced lining with long-lasting and non-combustible Frankosorb® absorbers
- › Specialized for ‘out-of-the-range’ EMC test environments
- › Usable for automotive and military standard tests
- › Turnkey solutions



SAC-10-3/H	18,380 x 12,830 x 8,550 m (L x W x H) QZ ø3,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-4/H	19,280 x 13,280 x 8,550 m (L x W x H) QZ ø4,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-5/H	21,080 x 15,080 x 8,700 m (L x W x H) QZ ø5,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-6/H	21,680 x 15,680 x 8,700 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000, H600, and H1300 Turbine absorbers
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (26/80 MHz to 18 GHz)

SAC-10/P Pyramid

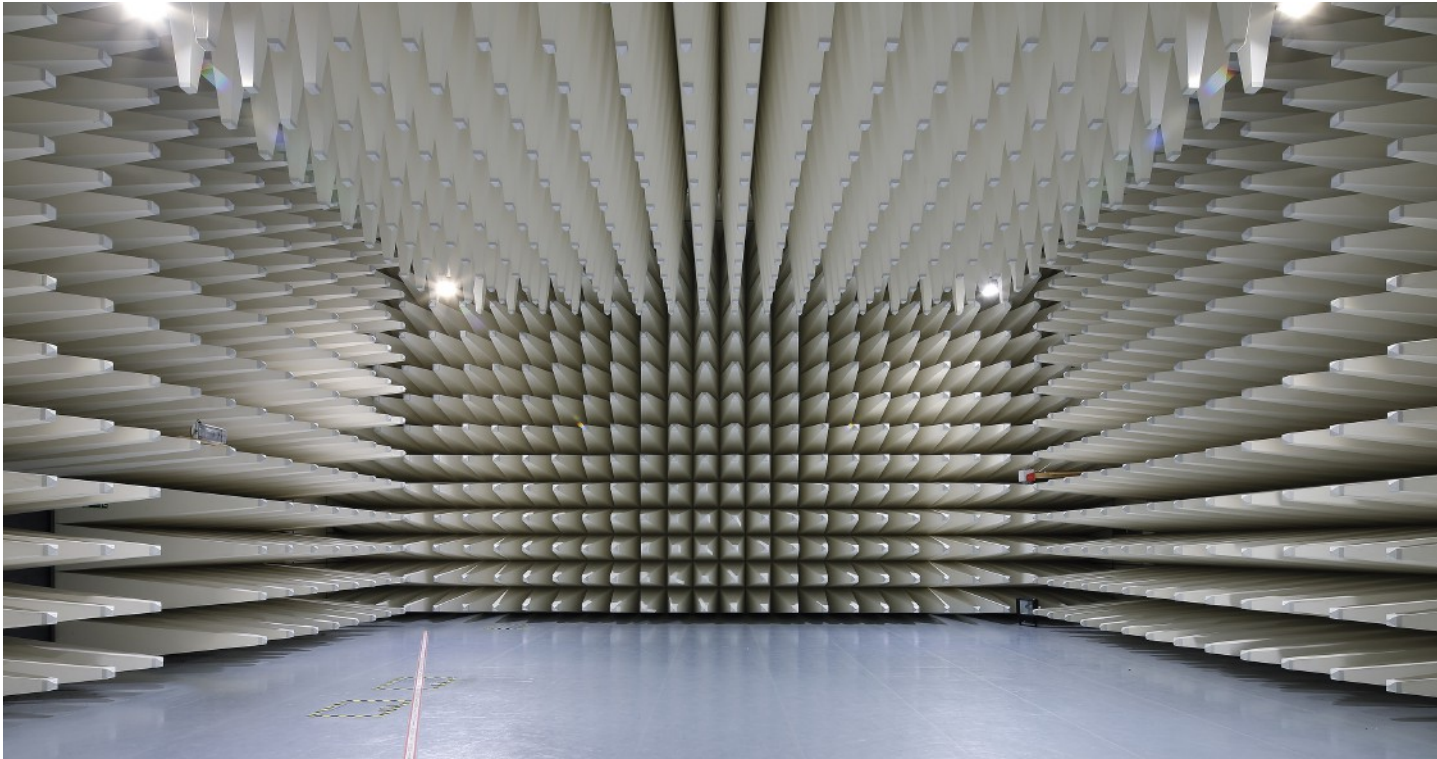
10m Semi Anechoic Chamber with Long-pyramid Absorbers

The **SAC-10/P** is Frankonia’s full compliant and customizable EMC testing solution at **10,0 m** measuring distance with a Quiet Zone (QZ) of **ø3,0 m** up to **ø6,0 m** and the unique Frankonia **long-pyramid** absorber layout.

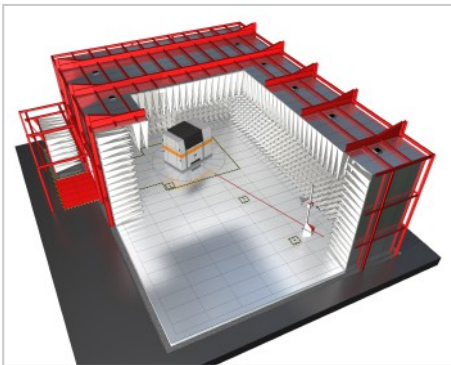
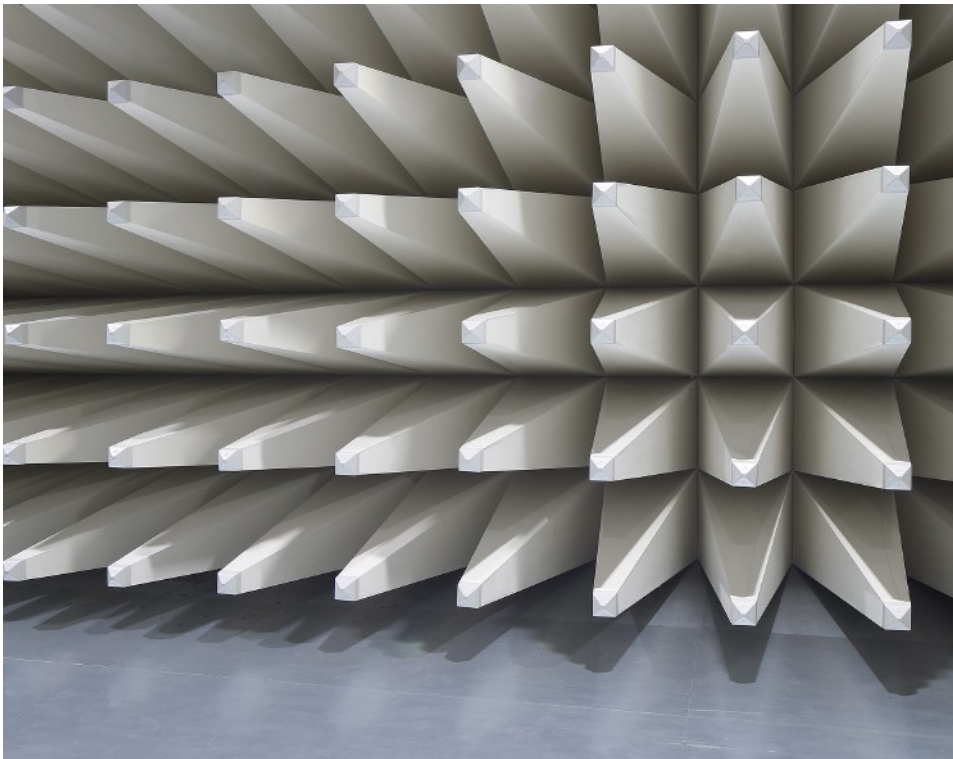
Due to the high grade of customization reflecting the demands of our customers, this semi-anechoic chamber is adaptable in size and offers several configuration possibilities.

Features

- › Full lining with long-lasting and non-combustible Frankosorb® long-pyramid absorbers (Frankonia technology)
- › Cost-efficient alternative to hybrid absorber lining without any limitations
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Full compliant with military and automotive standards
- › Highly customizable solution for any kind of EMC testing and limitless integration of individual applications; as single or double test axis option
- › Notably adjustable anechoic chamber size, characteristics and configuration due to different EUT requirements
- › Specialized for ‘out-of-the-range’ EMC test environments
- › Usable for automotive and military standard tests
- › Floor absorbers storage below the pyramids
- › Turnkey solutions



The innovative long-pyramid absorber technology achieves exceptional performance for emissions and immunity testing and offers the highest homogeneity and impedance accuracy for the complete frequency range.



SAC-10/P
Long-pyramid Absorbers

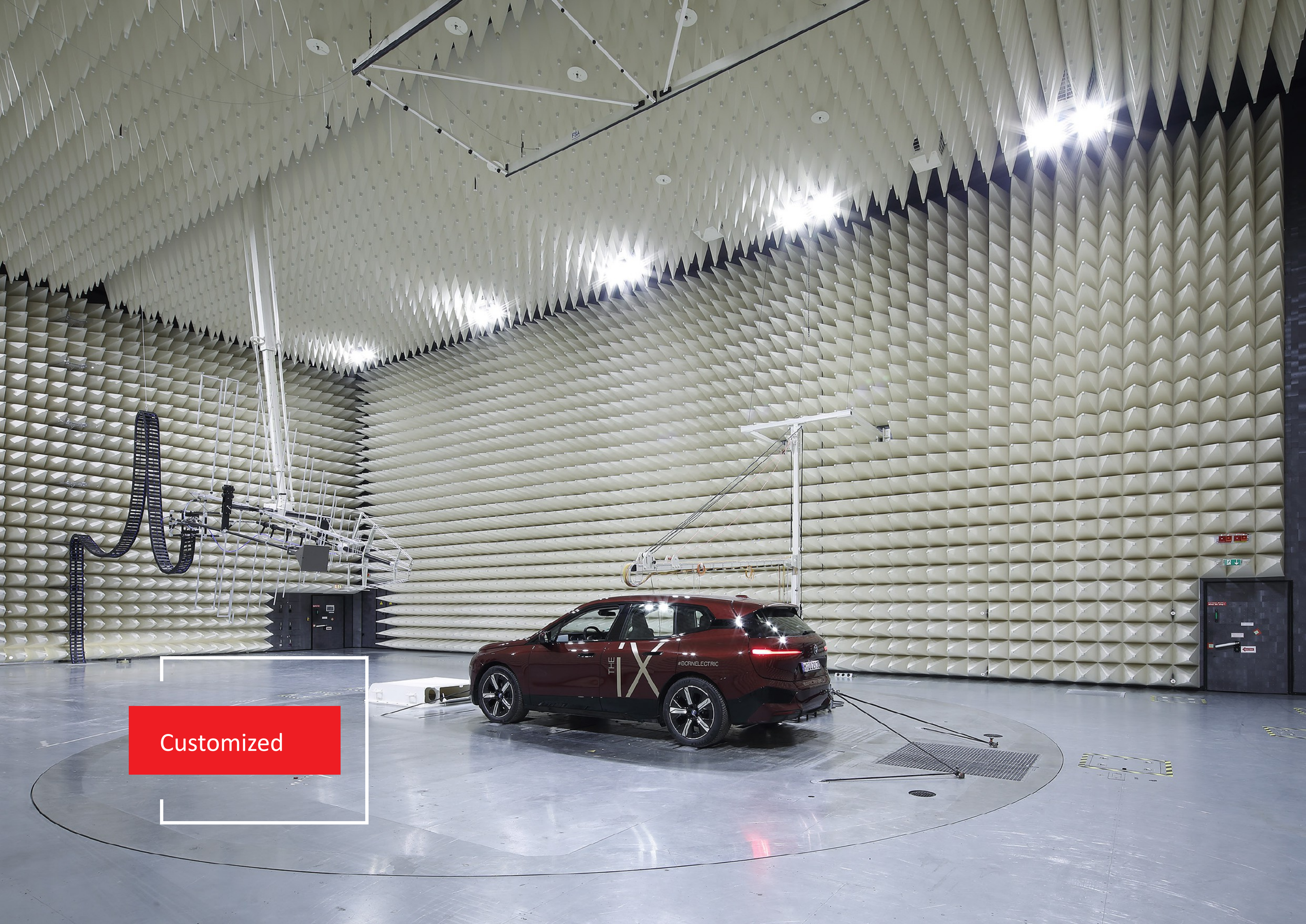
SAC-10-3/P	21,680 x 13,730 x 8,550 m (L x W x H) QZ ø3,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-4/P	21,680 x 13,730 x 8,550 m (L x W x H) QZ ø4,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-5/P	23,480 x 16,580 x 9,000 m (L x W x H) QZ ø5,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10-6/P	24,980 x 17,180 x 9,000 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m)
Frequency range	9 kHz/ 30 MHz to 18 GHz (option 40 GHz)

Absorbers

- › Frankosorb® long-pyramid absorber lining with P2400
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (26/80 MHz to 18 GHz)



Customized

ACTC

CISPR 25 Automotive Component Testing Chamber

The **ACTC** is Frankonia’s automotive component testing chamber solution at 1,0 m measuring distance. This chamber solution is adapted to full compliant tests of automotive components according to CISPR 25 and ISO 11452. A permanent plug-in contact strip is installed between the absorbers to ensure the electrical connection of the test table to the shielding, and includes the test table as required by CISPR 25.

The **UCC** is Frankonia’s ultra-compact hybrid solution at 1,0 m measuring distance. The chamber is designed for pre-compliance radiated emission and immunity tests, conducted tests, and pre-compliance tests for automotive components as per the CISPR 25 method. It is an alternative solution for the GTEM cell for pre-compliance testing as well as for research and scientific purposes in all sectors.

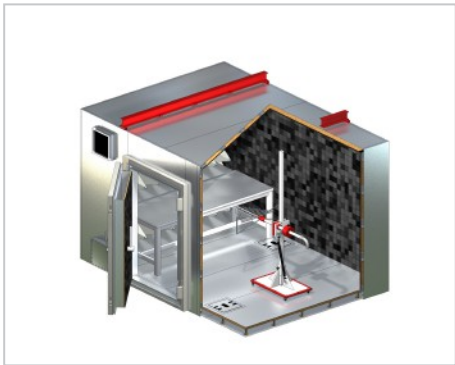
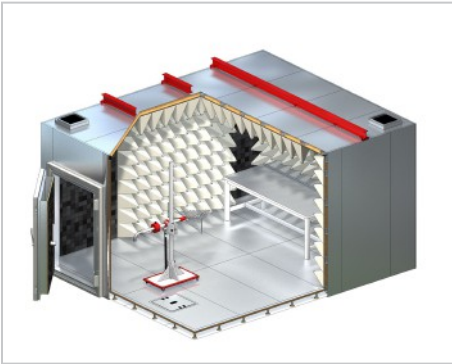
Features

- › **ACTC:** Full compliant per CISPR 25 and ISO 11452 for components
- › **ACTC L:** Full compliant per CISPR 25 and ISO 11452 for components and large enough for tests on vehicles
- › **UCC:** Pre-compliant per CISPR 25 and ISO 11452 (alternative to GTEM cell)
- › Compact chamber solution for automotive component testing
- › Upgradeable for E-Drive (load machine, BlueBox, battery test system)
- › Advanced and optimized lining with long-lasting Frankosorb® hybrid absorbers



ACTC

Compliant Automotive Component Testing



UCC

Pre-compliant Automotive Component Testing



ACTC	6,380 x 5,480 x 3,750 m (L x W x H) CISPR 25 component level at 1,0m test distance
ACTC L	11,480 x 6,580 x 4,500 m (L x W x H) CISPR 25 component level & vehicle at 1,0 m test distance
UCC	4,580 x 3,080 x 2,550 m (L x W x H) Pre-compl. component level at 1,0 m test distance
Frequency range	150 kHz/ 26 MHz to 18 GHz (option 40 GHz)

Performance & Compliance **ACTC**

- › Full compliant emission (EMI) according to CISPR 25
- › Full compliant immunity (EMS) according to ISO 11452
- › Compliant immunity (EMS) according to IEC/EN 61000-4-3
Uniform filed 0,5 x 0,5 m at 1,0 m measuring distance
Deviation FU 0/+6 dB at 100 % (26/80 MHz to 18 GHz)

Performance & Compliance **ACTC L**

- › Full compliant emission (EMI) according to CISPR 25
- › Full compliant immunity (EMS) according to ISO 11452
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Uniform field 1,5 x 1,5 m at 3,0 m measuring distance
Deviation FU 0/+6 dB at 75 % of 16 measuring points (26/80 MHz to 18 GHz)

Performance & Compliance **UCC**

- › Pre-compliant emission (EMI) according to CISPR 25
- › Pre-compliant immunity (EMS) according to ISO 11452

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite and H450
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

AVTC

3m Automotive Vehicle Testing Chamber

The **AVTC** is Frankonia’s automotive anechoic chamber solution at **3,0 m** or **5,0 m** measuring distance offering a Quiet Zone (QZ) of **ø4,0 m** for commercial testing combined with a focus on automotive component and vehicle tests.

It is adapted for radiated emissions on vehicles acc. to CISPR 12 and components acc. to CISPR 25 as well as for commercial product tests acc. to CISPR 16-1-4 and ANSI C63.4. Furthermore, it is adapted to radiated immunity acc. to IEC/EN 61000-4-3, ISO 11451 and ISO 11452.

Features

- › Advanced Frankosorb® hybrid absorber lining
- › Automotive component & vehicle and commercial tests in a single solution
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Full compliant with CISPR 25, CISPR 12, ISO 11452 and ISO 11451
- › ECE R10 with integrated or mobile dynamometer up to 3,0 m test distance
- › Cost-effective and high-performance solution for 3,0 m or 5,0 m test distance
- › Floor absorber board for an efficient and fast modification of the test setup
- › Upgradeable with EDTC components (load machine, BlueBox, etc.)
- › Highly customizable solution for any kind of EMC testing and limitless integration of individual applications
- › Usable for commercial and military standard tests



AVTC	11,480 x 9,380 x 6,000 m (L x W x H) QZ ø3,0 m at 3,0 m test distance (H= 2,5 m) e.g., with turntable up to ø5,0 m
AVTC L	14,780 x 11,480 x 6,300 m (L x W x H) QZ ø3,0 m at 3,0 m and 5,0 m test distance (H= 2,5 m) e.g., with turntable up to ø6,0 m
AVTC XL	16,280 x 12,680 x 6,300 m (L x W x H) QZ ø4,0 m at 3,0 m and 5,0 m test distance (H= 2,5 m) e.g., with integrated dynamometer ø7,0 m
Frequency range	9 kHz/ 150 kHz to 18 GHz (option 40 GHz)

Absorbers

- › Optimized Frankosorb® hybrid absorber lining with Ferrite, H1000 and H600
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
Deviation FU 0/+6 dB at 75 % of 16 measuring points (26/80 MHz to 18 GHz)
- › Full compliant emission (EMI) according to CISPR 25 and CISPR 12
- › Full compliant immunity (EMS) according to ISO 11452 and ISO 11451
- › ECE R10 at 3,0 m test distance with dynamometer

SAC-10V

10m Semi Anechoic Chamber for ECE R10 Vehicle Testing

The **SAC-10V** chamber is Frankonia’s full compliant and customizable EMC testing solution at **10,0 m** measuring distance offering various sizes of Quiet Zone (QZ) and are dedicated to automotive full vehicle testing with integrated dynamometer. Due to the high grade of customization reflecting the demands of our customers, this semi-anechoic chamber is adaptable in size and offers several configuration possibilities.

Features

- › **SAC-10V-6/H:** Optimized and advanced Frankosorb® hybrid absorber lining
- › **SAC-10V-6/P:** Full lining with Frankosorb® long-pyramid P2400 absorbers
- › Full compliant EMI acc. to CISPR 16-1-4 and ANSI C63.4
- › Full compliant EMS acc. to IEC/EN 61000-4-3
- › Full compliant with CISPR 25, CISPR 12, CISPR 36, ISO 11452 and ISO 11451
- › ECE R10 with integrated dynamometer at 10,0 m test distance
- › Highly customizable for any kind of testing, vehicle sizes and weights, limitless integration of applications
- › Specialized for ‘out-of-the-range’ EMC test environments
- › Usable for commercial and military standard tests

Absorbers

- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)



Performance & Compliance

- › Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
- › Full compliant emission (EMI) according to CISPR 25 and CISPR 12
- › Full compliant immunity (EMS) according to IEC/EN 61000-4-3
- › Full compliant immunity (EMS) according to ISO 11452 and ISO 11451
- › ECE R10 at 10,0 m test distance with dynamometer



SAC-10V
Vehicle Test Chamber
with Dynamometer

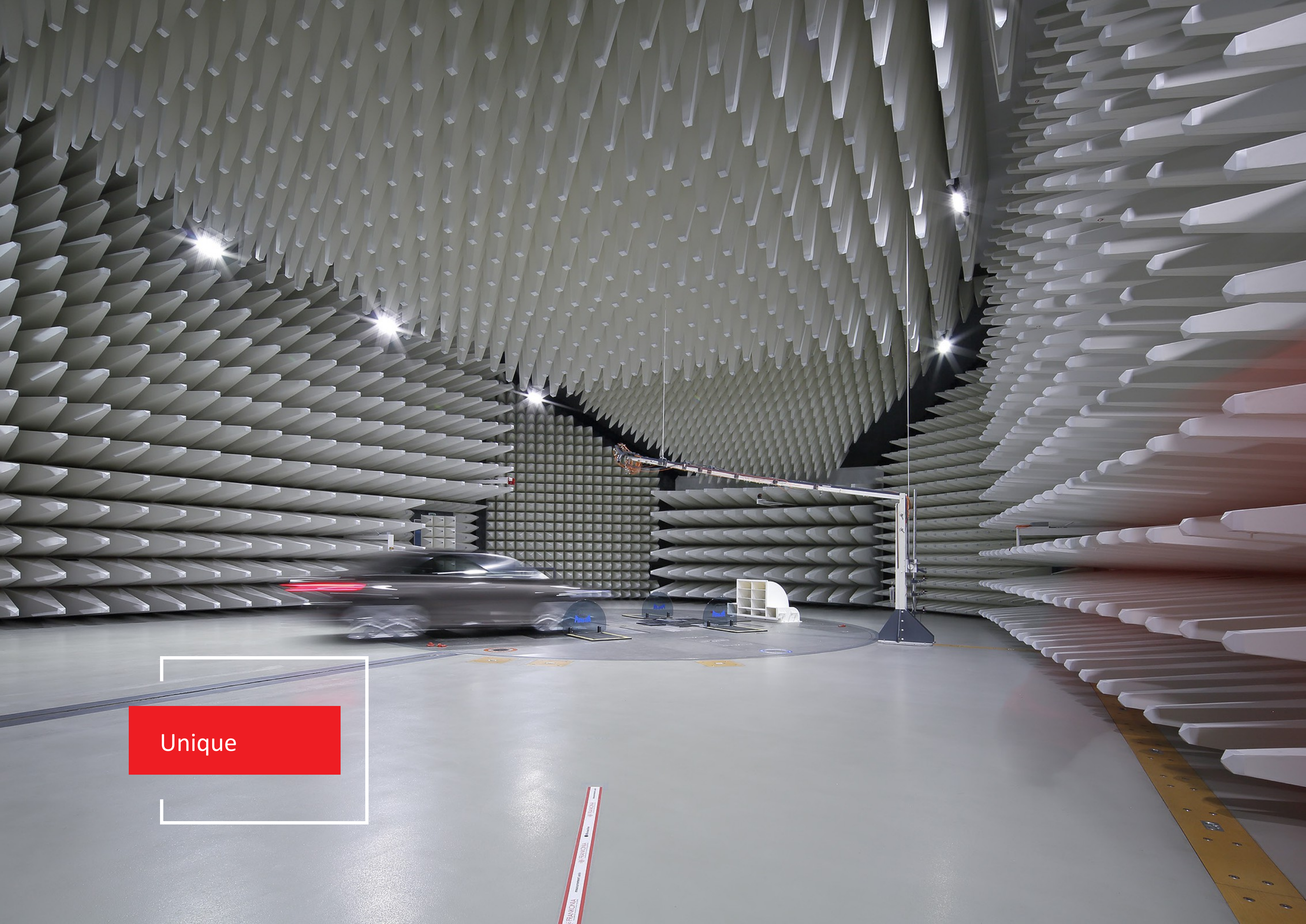


Hybrid Absorber Solution

SAC-10VC-6/H	23,030 x 14,480 x 6,300 m (L x W x H) QZ ø6,0 m at 5,0 m test distance (H= 2,5 m) Prepared for a 10,0 m test distance for vehicle tests
SAC-10V-6/H	22,580 x 15,680 x 8,700 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10V-6/H (SL12)	24,380 x 16,580 x 9,000 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m) Heavy load test zone up to 12 m long vehicles
SAC-10V-6/H (SL18)	26,780 x 18,080 x 9,000 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m) Heavy load test zone up to 18 m long vehicles
Frequency range	9 kHz/ 150 kHz to 18 GHz (option 40 GHz)

Pyramid Absorber Solution

SAC-10V-6/P	26,480 x 20,180 x 9,000 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m)
SAC-10V-6/P (SL12)	26,480 x 20,180 x 10,500 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m) Heavy load test zone up to 12 m long vehicles
SAC-10V-6/P (SL18)	30,080 x 20,180 x 10,500 m (L x W x H) QZ ø6,0 m at 10,0 m test distance (H= 3,0 m) Heavy load test zone up to 18 m long vehicles
Frequency range	9 kHz/ 150 kHz to 18 GHz (option 40 GHz)



Unique

EDTC & BlueBox

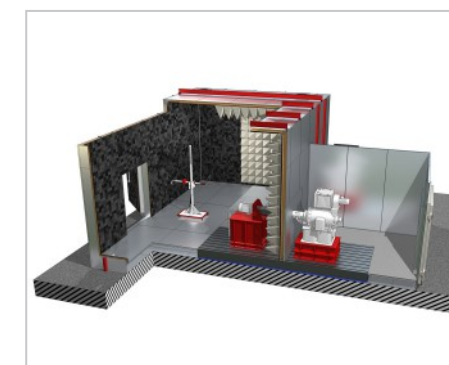
E-Drive Test Solutions with External Load Machines or Mobile Load Machine

E-Drive Test Solutions are Frankonia's dedicated test sites for powertrain components and facilities related to hybrid, electric, fuel cell and battery drive systems. They offer superior conditions for radiation testing according to CISPR 25 and ISO 11452.

The **EDTC-BB** is the adapted chamber solution that includes the **EMC-BlueBox** mobile load machine for dynamic EMC tests of electrical powertrain units in a shielded enclosure. The BlueBox works in a four-quadrant operation; any EUT stress situation can be simulated. Similar to the external load machine with a fixed shaft, it includes braking, driving, direction of rotation (right/left), speed regulation, torque control and a mix out of this range.

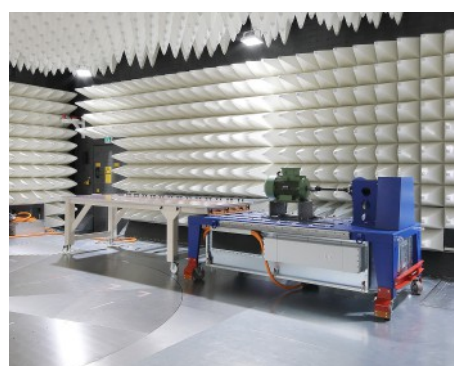
Features

- › Fully compliant with CISPR 25 and ISO 11452
- › Optimized Frankosorb® hybrid absorber lining
- › Component or system test level
- › Mobile, flexible and adjustable to any kind of EUT
- › 360° view when placed on a turntable (extended testing range)
- › Combination with battery tests
- › Integration kit for existing chambers
- › Optional EUT e-motor power source and water cooling system
- › Easy to use for every EMC expert



EDTC-SA
External Load Machine
for E-Motor Tests

EMC-BlueBox
Mobile Load Machine

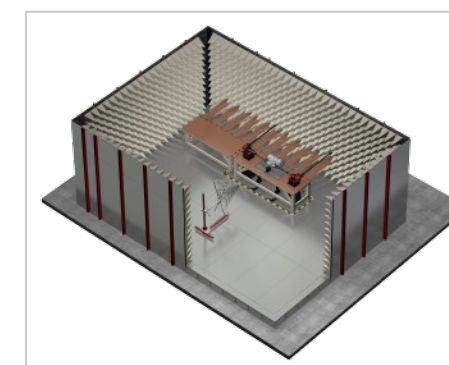


The **EDTC-SA** is the chamber solution that is specifically prepared for a **single external load machine** with fixed shaft. The **EDTC-AX** is the chamber setup that is defined for e-axle test on powertrain units and is prepared for **two external load machines** with fixed shaft. The patented system can be used for dynamic drive tests of electrical powertrain units and matches with your preferred dynamometer supplier. Frankonia focus to the proper EMC setup inside an EMC chamber and provides adapted test tables, grounding conception, and 90° angle gear boxes, and is open to any dynamometers.

EDTC-SA	7,880 x 5,480 x 3,750 m (L x W x H) for fixed-shaft version with external load machine e.g., 1x 250 kW with 3.000 RPM and 3.000 Nm
EDTC-AX	9,080 x 6,080 x 3,750 m (L x W x H) for fixed-shaft version with external load machines e.g., 2x 250 kW with 3.000 RPM and 3.000 Nm
EDTC-BB	7,880 x 6,380 x 3,750 m (L x W x H) for mobile load machine EMC-BlueBox up to 120 kW
EDTC-BB with turntable	10,880 x 6,980 x 3,900 m (L x W x H) for mobile load machine EMC-BlueBox up to 120 kW with turntable for 360° scan
EDTC-HY	6,380 x 5,480 x 3,750 m (L x W x H) for hydraulic load machine, e.g., 2x 250 kW

Features

- › Motor adapter, grounding and connection per CISPR 25
- › Vibration-free and non-interacting solid basement (floating slab)
- › Extended services in consultancy and test readiness guidance



EDTC-AX
External Load Machines
for E-Axle Tests

External Load Machine (example)

Type	EDTC-250	EDTC-500
Power	1x250 kW	2x250 kW
Speed	3.000 RPM	3.000 RPM
Torque	3.000 Nm	3.000 Nm

Mobile Load Machine EMC-BlueBox

Type	BlueBox-30	BlueBox-40	BlueBox-65	BlueBox-120
Power	30 kW	40 kW	63 kW	120 kW
Speed	11.000 RPM	9.000 RPM	6.500 RPM	6.000 RPM
Torque	82 Nm	140 Nm	240 Nm	470 Nm

Reverberation Chambers

Reverberation Solutions with Frankonia Performance Stirrers

The RVC Chambers are designed based on Frankonia’s modular construction system. Prefabricated high-quality shielding panels (>8 MS/m) guarantee a maximum of flexibility and performance. The system offers a high grade of customization reflecting the demands of our customers. The shielding panels can be installed reverse (flat surface inside) or as regular PAN shielding with mounting from the inside, which allows future upgrades, e.g., absorber lining.

Features & Compliance

- › Retrofit of a RVC with Frankosorb® hybrid absorbers and regular PAN shielding or converting old EMC chambers to RVC chamber
- › Cost-effective and performance solution for small products up to vehicles
- › Immunity compliance according to IEC/EN 61000-4-21 and ISO 11452-11
- › Immunity and emission compliance according to ISO 11451-5 (fast stirring)
- › Full safety integration per Machinery Directive (2006/42/EC)

Frankonia Stirrers

- › Regular Z-Fold Stirrers with up to 30 RPM at e.g. ø1,8 m
- › Performance Z-Fold Stirrers with up to 60 RPM at e.g. ø2,8 m
- › Disc-style Stirrers with up to 120 RPM at e.g., ø4,0 m
- › Tube-style Stirrers with up to 240 RPM at e.g., ø2,0 m
- › Large-disc Stirrers with up to 10 RPM at e.g., ø12,0 m

Frankonia offers various stirrer designs and concepts, and maintains its role as a solution provider by adapting our customers stirrer designs, which we implement then in new RVC chambers or converting old chambers.

Commercial & Industrial RVC

RVC e1	7,580 x 5,630 x 4,200 m (L x W x H) Working volume: 3,3 x 3,5 x 2,6 m (L x W x H) LUF: 200 MHz Stirrer: 1x Z-Fold Stirrer (vertical oriented) Products: Small or medium size ISM & multimedia
RVC e2	11,280 x 7,280 x 4,950 m (L x W x H) Working volume: 5,5 x 4,0 x 2,6 m (L x W x H) LUF: 80 MHz Stirrer: 2x Z-Fold Stirrer (vertical and horizontal oriented) Products: Large ISM & multimedia

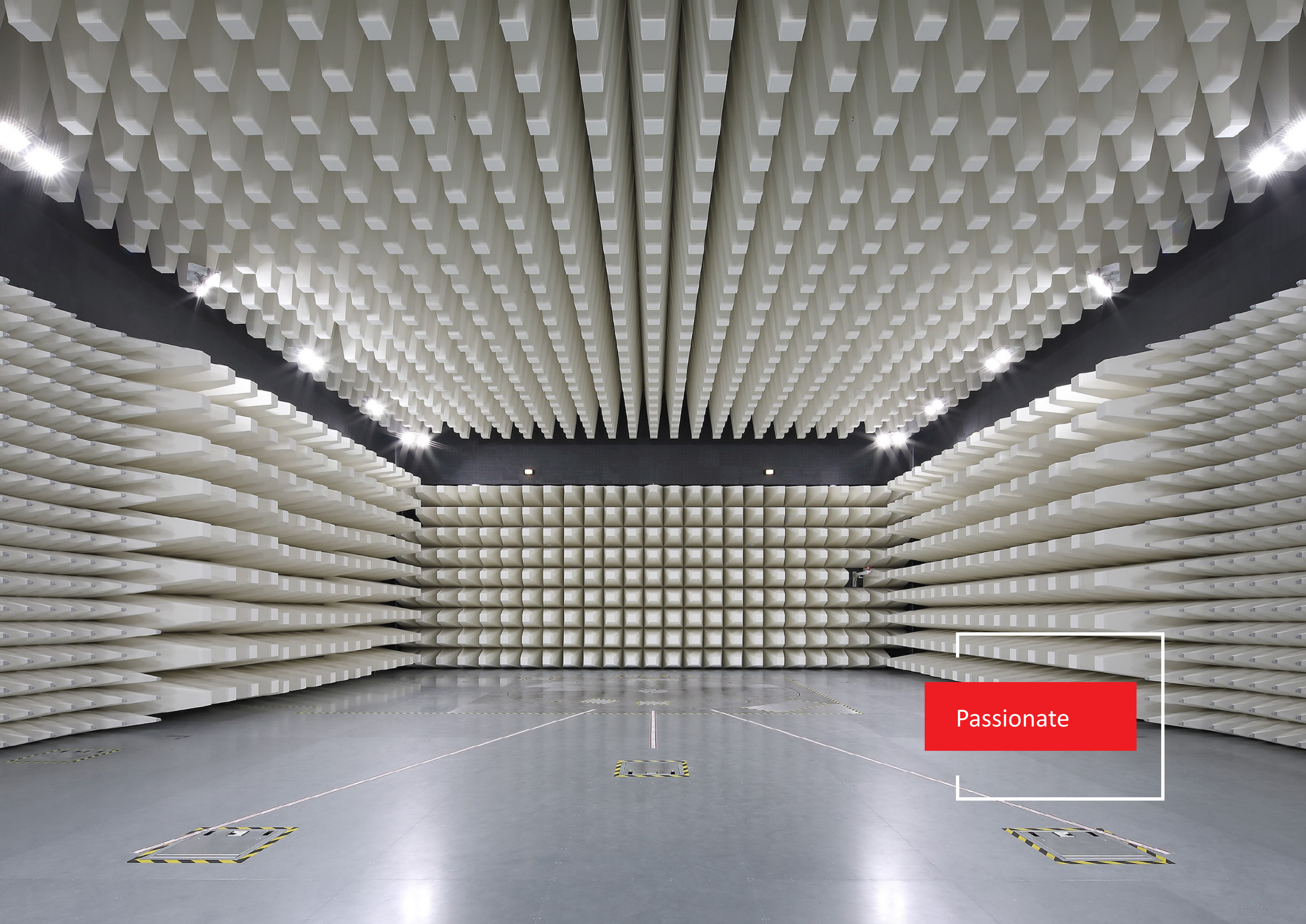


Automotive RVC

RVC S	5,330 x 3,380 x 3,300 m (L x W x H) Working volume: 2,5 x 1,0 x 1,5 m (L x W x H) LUF: 200 MHz Stirrer: 1x Z-Fold Stirrer (vertical oriented) Products: Components for Military or Automotive
RVC M	7,580 x 5,630 x 4,200 m (L x W x H) Working volume: 3,3 x 3,5 x 2,6 m (L x W x H) LUF: 200 MHz Stirrer: 1x Z-Fold Stirrer (vertical oriented) Products: Large components for Military or Automotive
RVC L	13,880 x 11,480 x 6,300 m (L x W x H) (custom) Working volume: 8,0 x 5,0 x 3,0 m (L x W x H) LUF: 80 MHz Stirrer: 2x Z-Fold Stirrer (vertical and horizontal oriented) Products: Vehicles
RVC XL	15,530 x 11,480 x 6,600 m (L x W x H) (custom) Working volume: 8,0 x 5,0 x 3,0 m (L x W x H) LUF: 80 MHz Stirrer: 1x Large-disc Stirrer ø9,0 m, 2x Disc Stirrer ø4,0 m Products: Vehicles
RVC XXL	17,480 x 13,580 x 6,600 m (L x W x H) (custom) Working volume: 8,0 x 5,0 x 3,0 m (L x W x H) LUF: 80 MHz Stirrer: 1x Large-disc Stirrer ø12,0 m, 2x Disc Stirrer ø4,0 m Products: Large vehicles



RVC
Reverberation Chambers



Passionate

Military Chamber Solutions

Anechoic Chambers for Military Applications

The **MIL-STD Chamber** is Frankonia's large chamber solution at **1,0 m** measuring distance according to MIL-STD 461 adapted for radiated emission and immunity tests for large EUT's or vehicles.

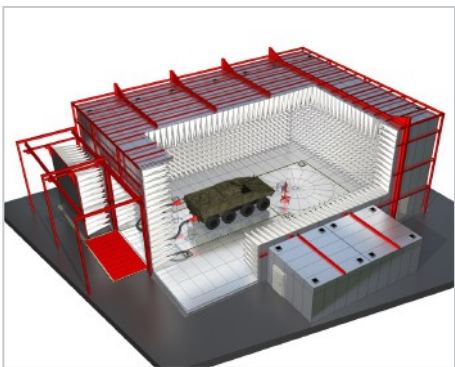
- › Full compliant acc. to MIL-STD 461 and DO-160
- › Frequency of absorbers: 80 MHz to 40 GHz

The **MIL-STD Advanced Chamber** is Frankonia's military chamber solution acc. to MIL-STD 461 for large EUT's, and is compliant with commercial or automotive test site requirements. Both solutions are designed based on MIL-STD 461 and are fully customized according to customers' requirements for military testing of large and heavyweight EUT's.

- › Full compliant acc. to MIL-STD 461 and DO-160
- › Full compliant EMI/EMS for commercial and automotive standards
- › Frequency of absorbers: 30 MHz to 40 GHz

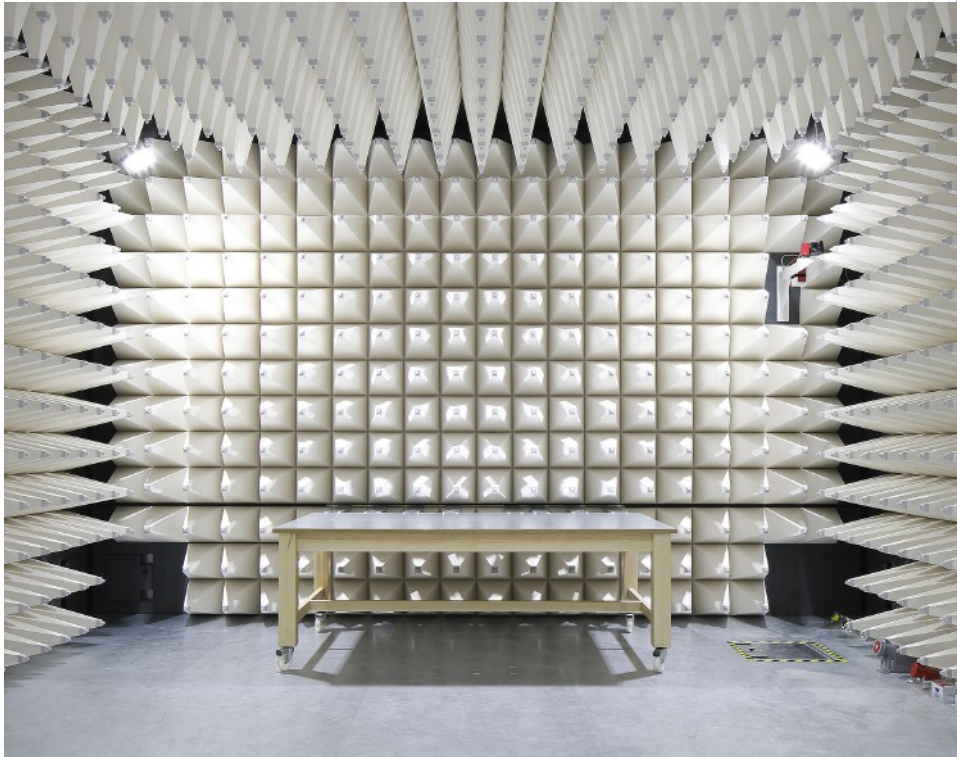
The **MIL CHC** ist Frankonia's Compact Hybrid Chamber, lined with Frankosorb® Hybrid Absorber Layout. The **MIL CPC** is Frankonia's Compact Parymid Chamber, lined with Frankosorb® Short Pyramid Absorbers. These chamber solutions are adapted for full compliant radiated emission and immunity tests of lightweight EUT's acc. to MIL-STD 461 and DO-160 at 1,0 m test distance.

- › Full compliant for components acc. to MIL-STD 461 and DO-160



MIL-STD Chamber
Military Equipment, Vehicles,
Commercial and Automotive
30 MHz to 40 GHz

MIL CHC & MIL CPC
Military Component Testing



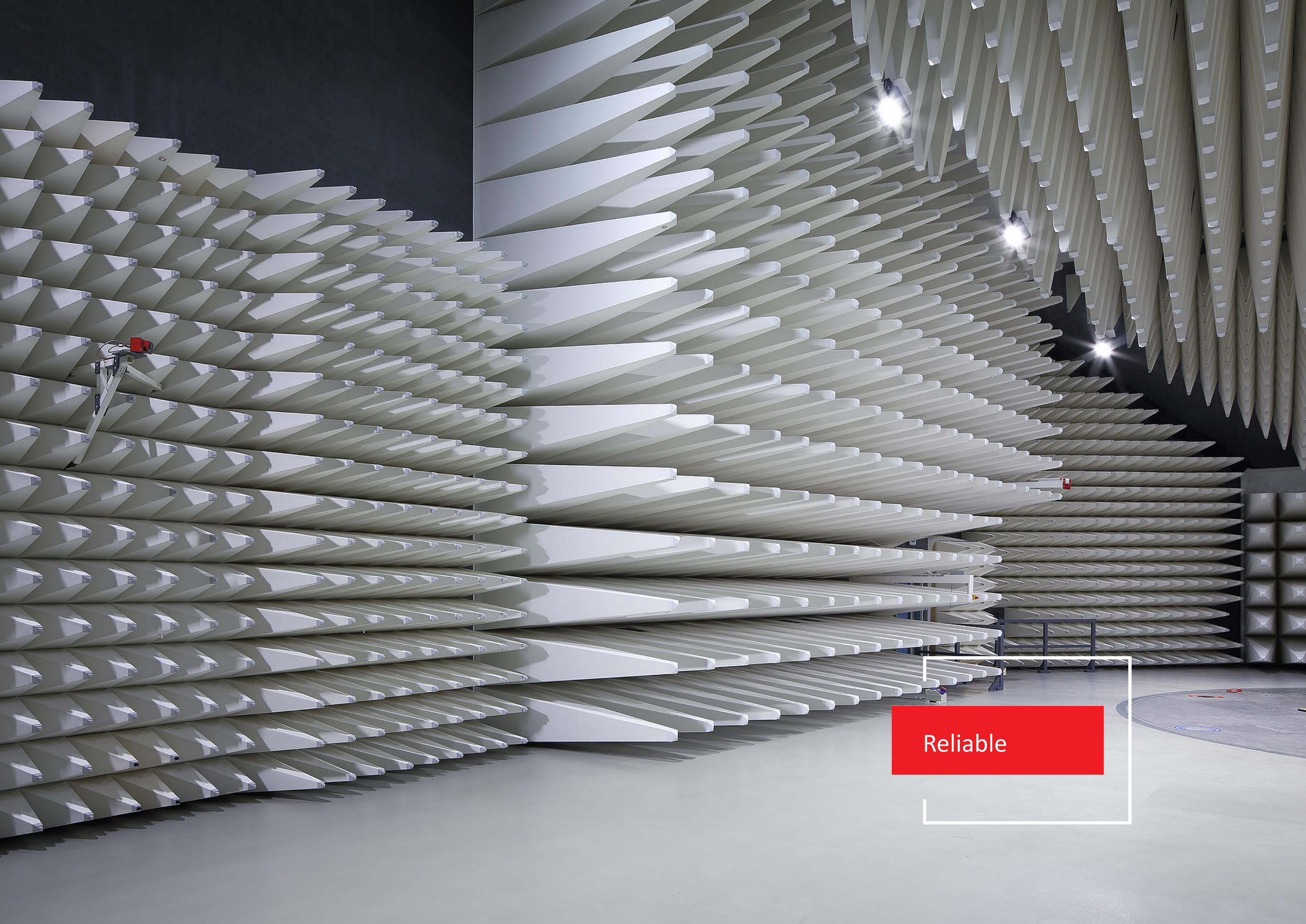
Absorbers

- › Frankosorb® short-pyramid, long-pyramid or hybrid absorber lining
- › High-performance nano thin-film technology with proven long-term stability
- › Non-combustible acc. to EN 13501-1 class A2 - s1 d0
- › Hardly inflammable acc. to EN 13501-1 class B (alternative)

MIL-STD Chamber	Custom size 9 kHz/ 80 MHz to 40 GHz with short-pyramid absorbers Feature: Military compliance
MIL-STD Advanced Pyramid	Custom size 9 kHz/ 26 MHz to 40 GHz with long-pyramid absorbers Feature: Military, industrial and automotive compliance
MIL-STD Advanced Hybrid	Custom size 9 kHz/ 30 MHz to 40 GHz with hybrid absorber lining Feature: Military, industrial and automotive compliance
MIL CHC	4,880 x 4,880 x 3,000 m (L x W x H) 9 kHz/ 30 MHz to 40 GHz with hybrid absorber lining
MIL CHC / DO-160	5,330 x 4,880 x 3,000 m (L x W x H) 9 kHz/ 30 MHz to 40 GHz with hybrid absorber lining
MIL CPC	6,080 x 5,380 x 3,750 m (L x W x H) 9 kHz/ 80 MHz to 40 GHz with short-pyramid lining

Performance & Compliance

- › Full compliant emission (EMI) & immunity (EMS) acc. to MIL-STD 461 and DO-160
- › Absorption at normal incidence:
 - 80 MHz to 250 MHz ≥ 6 dB, as per standard requirements
 - above 250 MHz ≥ 10 dB, as per standard requirements
- › Commerical compliance for MIL-STD Advanced
 - Full compliant emission (EMI) according to CISPR 16-1-4 and ANSI C63.4
 - Deviation NSA ±3,5 dB (30 MHz to 1 GHz)
 - Deviation SVSWR +5,5 dB (1 GHz to 18 GHz)
 - Deviation NSIL ±4,0 dB (9 kHz to 30 MHz)
 - Full compliant immunity (EMS) according to IEC/EN 61000-4-3
 - Deviation FU 0/+6 dB at 75 % of 16 measuring points (26/80 MHz to 18 GHz)



Reliable

Frankosorb®

Unique Nano Thin-film Absorbers

Since Frankonia's **Frankosorb®** nano thin-film absorber technology started to conquer the world market, more and more customers recognize and appreciate the stable performance characteristics and many other unique attributes offered by this technology. Frankosorb® convinces with more than 35 years of operation without having any malfunction, defect, quality or performance loss or the need to refurbish.

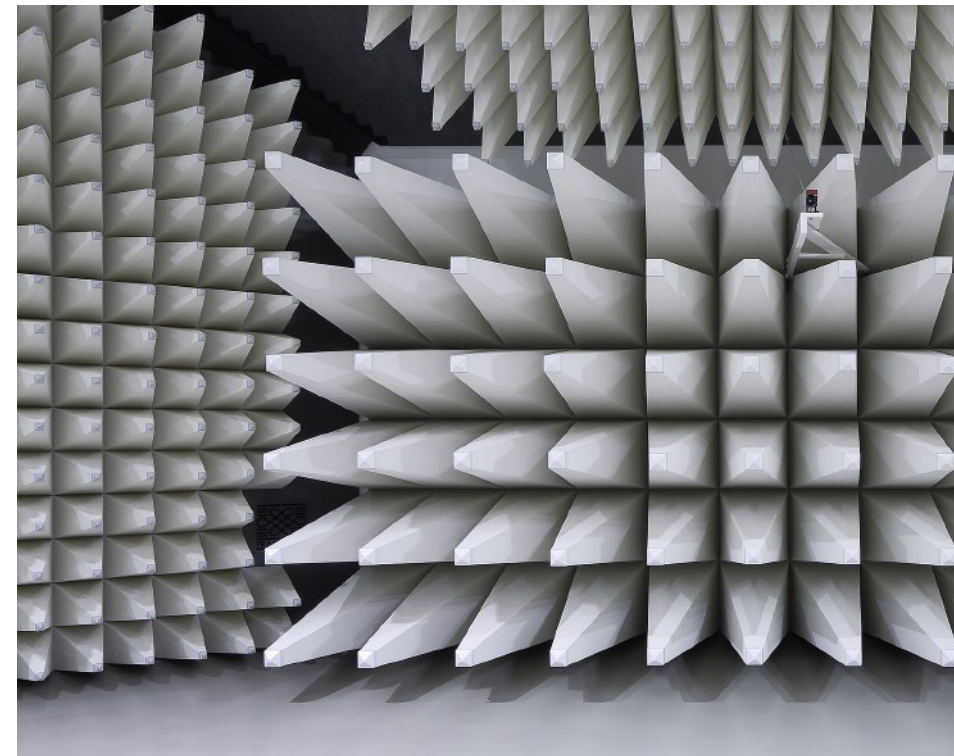
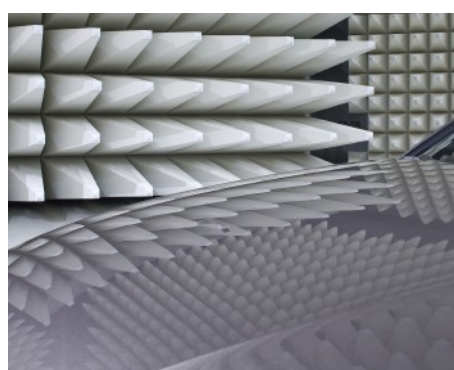
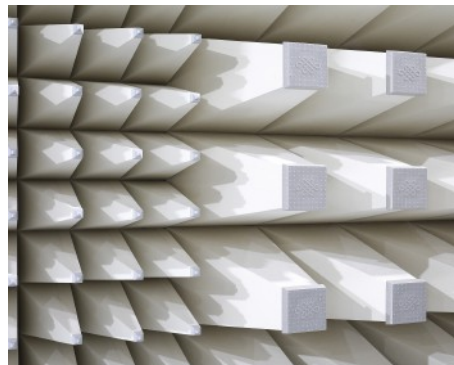
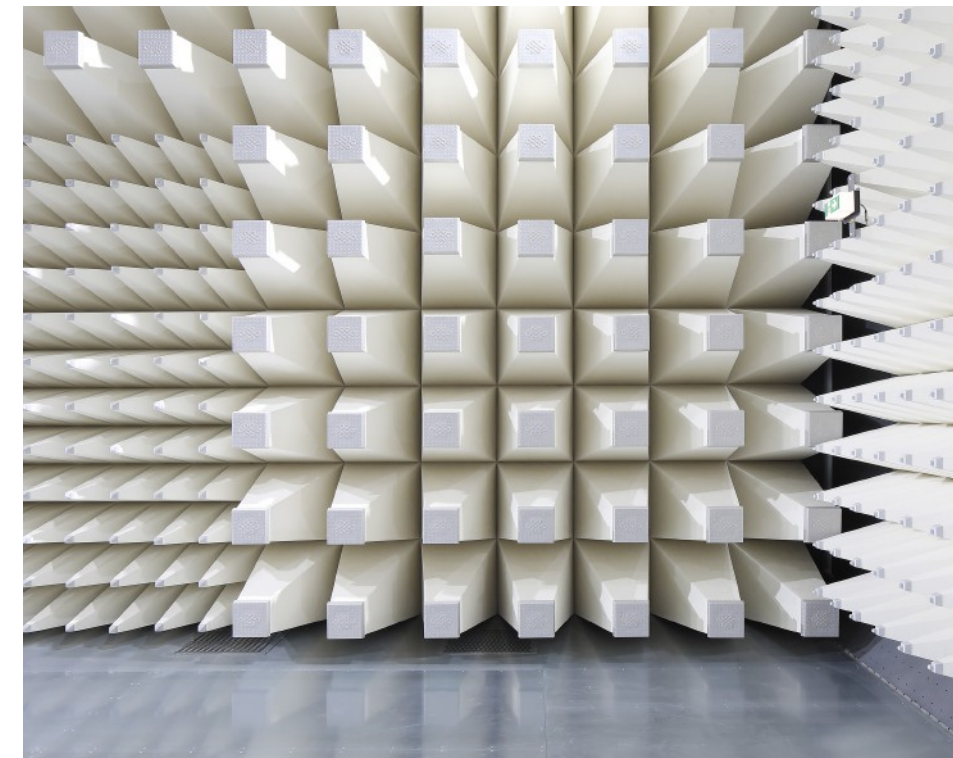
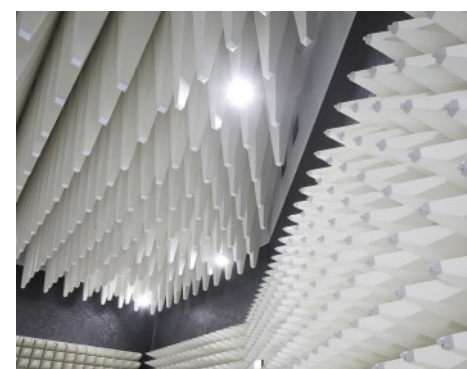
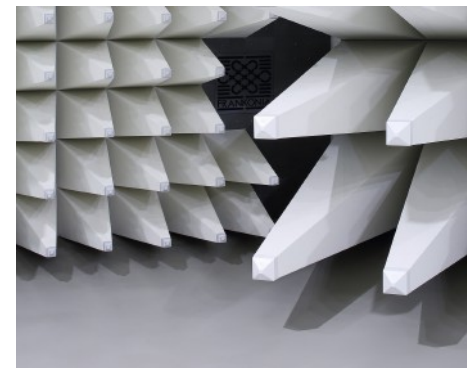
The unique method of Frankosorb® technology and its specific manufacturing process is available either as a hybrid solution in combination with Ferrite absorbers, or as a stand-alone pyramid solution with a length up to 2,4 m. The most important advantage of the Frankosorb® long pyramid absorbers is the coverage of the whole frequency range starting from 26 MHz up to 40 GHz, so that additional Ferrite absorbers are unnecessary and is thus a pure cost-saver.

Pyramid (P) Absorber Series

- › Short-pyramid absorbers from 80 MHz to 18/40 GHz: P600 or P900
- › Long-pyramid absorbers from 26 MHz to 18/40 GHz: P2000, P2200, or P2400

Hybrid (H) Absorber Series

- › Ferrite absorbers from 30 MHz to 1 GHz
- › Hybrid absorbers from 30 MHz to 18/40 GHz: H450, H600, or H1000
- › Performance Hybrid absorber from 30 MHz to 18/40 GHz: H1300 Turbine



Unique Features

- › High absorption capability paired with a fast cooling feature (hollow absorber)
- › Manufacturing process guarantees identical performances
- › No aging or drooping, no loss of performance, proven long-term stability for more than 35 years
- › White coloring improves the illumination level (no covers necessary)
- › Equal performances for Hybrid Absorbers and Long-pyramid Absorbers
- › Compliant with all existing verification standards, including magnetic field standards

People safety and laboratory protection

- › Preferred option that offers the safest environment:
Non-combustible absorbers according to DIN EN 13501-1 class **A2** - s1 d0 equivalent to DIN 4102 class A2, US NRL 8093 Tests 1,2,3,4 and 5; Chinese GB8624-2006; Russia GOST 30244-94; EN/ISO 5659-2 (smoke generation and opacity)
 Very high-power handling capacity up to 1 kW/m² or 600 V/m (continuous duty); 2,0 kW/m² or 850 V/m (intermediate power)
- › Alternative option:
Hardly inflammable absorbers according to DIN EN 13501-1 class **B** equivalent to DIN 4102 class B1, US NRL 8093 Tests 1,2 and 3; Chinese GB8624-2006; Russia GOST 30244-94
 Very high-power handling capacity up to 0,9 kW/m² or 550 V/m (continuous duty); 1,8 kW/m² or 800 V/m (intermediate power)
- › Cost protective solution with Frankosorb® non-combustible absorbers as no sprinkler or fire extinguishing system is necessary

Eco and user-friendliness

- › No toxic gases emitted in case of absorber heating
- › No dirt, no carbon dust, solvent-free, and free of glue or other harmful substances
- › Healthy environment for people and EUT
- › Recyclable at 99%
- › Non-hygroscopic materials are used to meet any climatic conditions (humidity-proof and temperature-proof)
- › Clean room classification according to ISO 14644-1
- › Easy to clean and washable; virus and bacteria resistant surface

The Frankosorb® absorber technology remains the number one choice when it comes to long-term performance paired with its unique non-combustibility characteristic.

Benefits

- › The Frankosorb® absorbers are available in several configurations that combines cost-effectiveness and high-performance. Thus, Frankonia's chambers offer the best choice for a long-term investment.
- › Due to the stable performance characteristics and its unique non-combustible attribute, a safe environment for people and EUT can be assured, which also leads to a constant, reproducible and long-lasting testing quality.
- › The Frankonia Frankosorb® absorber technology combines a variety of high-performance standards in a single solution.



Projected

Shielding & Accessories

Shielding, Doors, Gates, and Accessories

Since 1987, Frankonia follows a prefabrication and modular standard at highest quality and efficiency. Nothing is welded, nothing is glued, everything remains modular to meet any future modification requirements.

Shielding and Structure

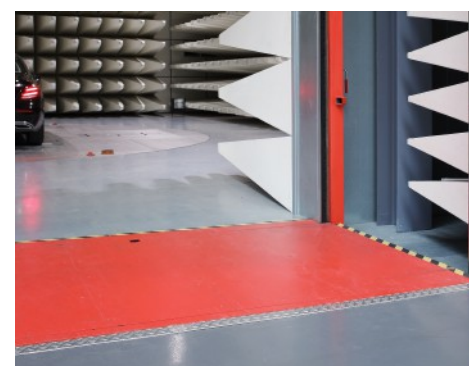
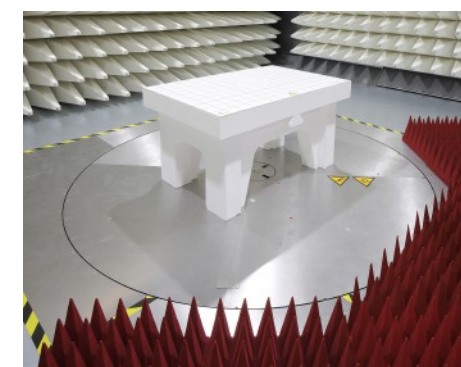
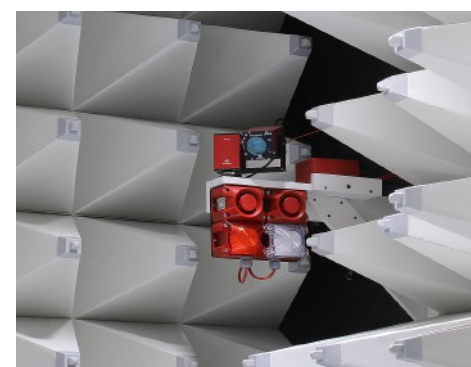
- › Modular and prefabricated PAN type shielding system
- › Highest shielding attenuation for every shielding accessories, honeycombs, doors, feed-through elements, electrics, and gates
- › Acoustic panels (FAP) absorption per ISO 354
- › Static steel structure adapted to local seismic conditions

Doors and Gates

- › Broad range of doors and gates in various sizes
- › Modular and prefabricated system
- › Single-leaf door (SLD)
- › Double-leaf door (DLD)
- › Sliding door (SSD)
- › Sliding gate (SG)

Ramps and Platforms

- › Automatic ramps
- › Automatic platforms (flush entrance)
- › Customized entrance solutions



As a specialist in RF-shielding and EMC testing chambers, Frankonia offers a broad range of complementary, standardized and customized products to maintain its position as a turnkey provider.

Every chamber is designed as an independent room with its own full-integrative electrical system setup, simple interfaces to the facility, with additional and individual accessories.

Electrical Integration and Compliance

- › Electrical distribution unit (outside accessible), cabling, safety functions per local standards
- › LED lighting, explosion protection option, emergency lighting
- › AC and DC filters, signal and data filters, optic converters
- › Safety matrix & higher-level laboratory control unit (PLC system)
- › CE conformity as per Machinery Directive 2006/42/EG for product compliance as standard, or for the complete laboratory as option, paired with advanced safety measures

Ventilation, Smoke and Gas

- › Honeycombs, cooling and exhaust systems
- › Air sampling network for gas and smoke detection systems
- › Smoke and gas detection analyzer with alarm central; ATEX compliance
- › Liquid detection system
- › Extinguishing solutions (Sprinklers, etc.)

Video & Audio Systems

- › HD camera systems, fixed or mobile version
- › Audio systems
- › Recording systems

Test Tables

- › CISPR 25 and MIL wooden test tables (FGT) with groundplane
- › CISPR 32 transparent test tables (FTT)

Automation

Turntables and Antenna Masts

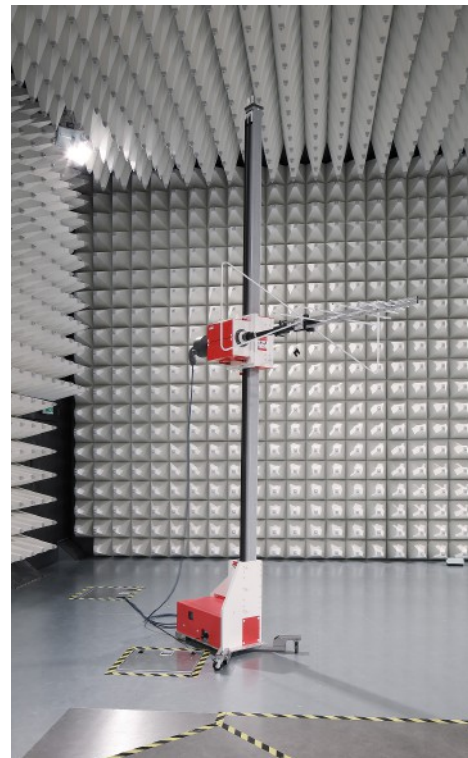
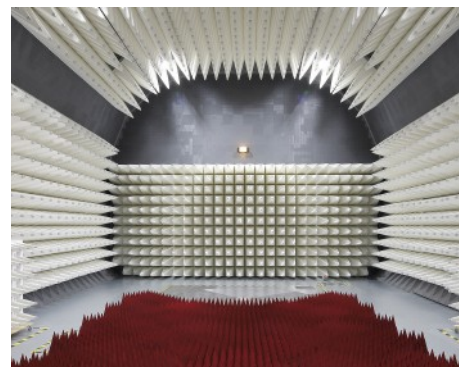
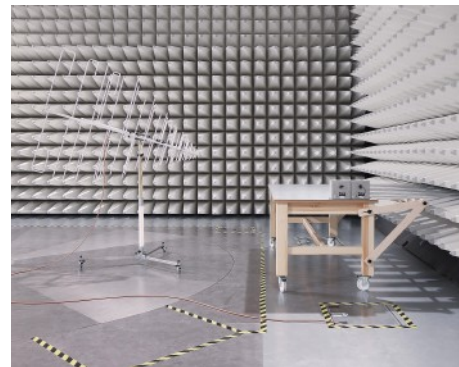
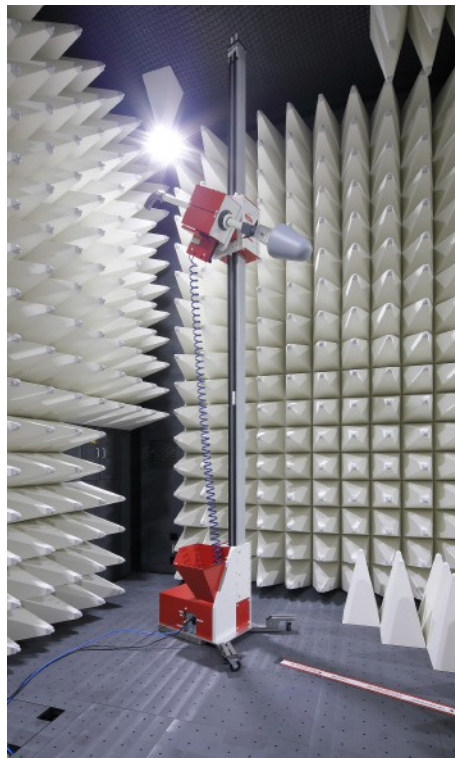
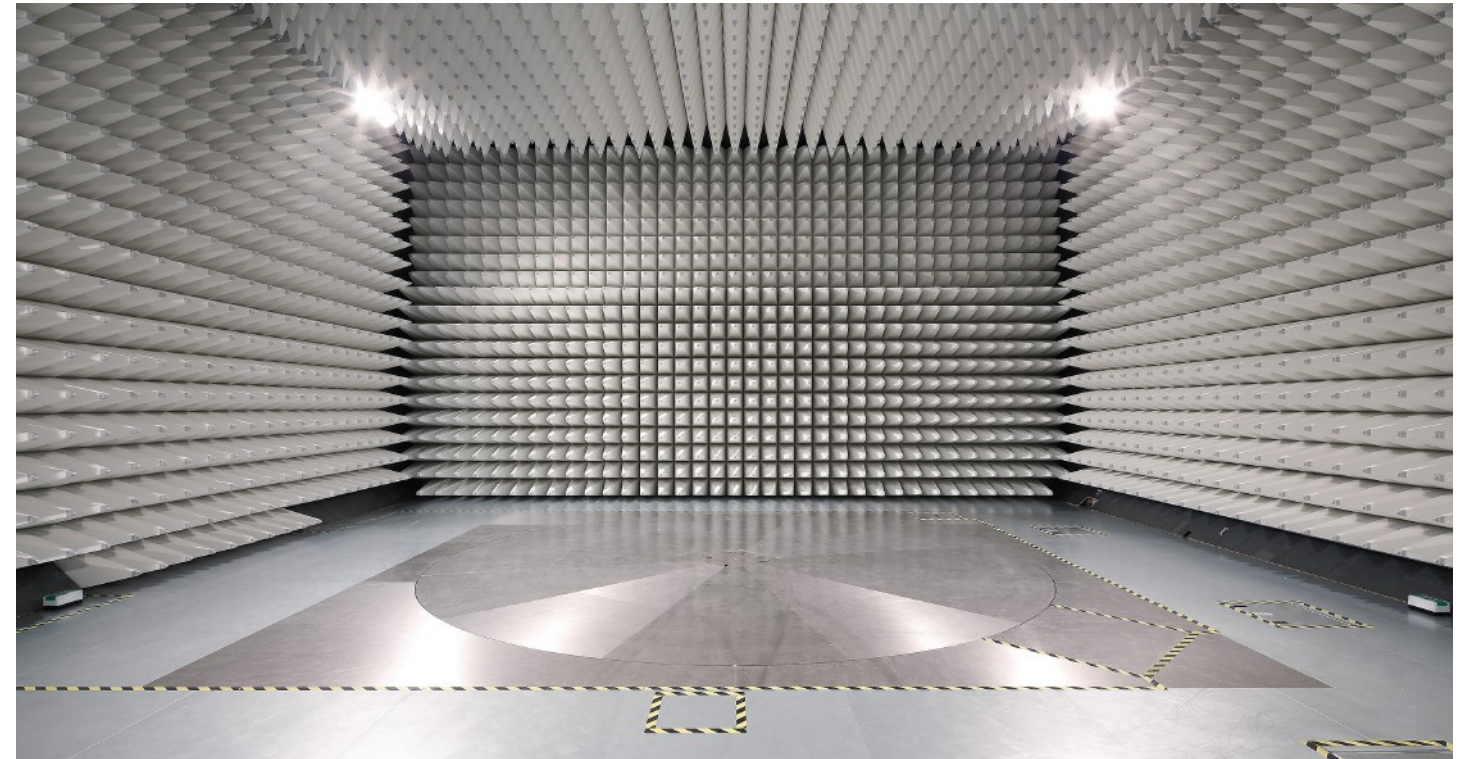
Frankonia provides a broad range of positioning devices such as standardized and multi-use turntable systems and antenna masts, which are designed and developed by Frankonia's own R&D department. Frankonia considers highest quality and technology standards respecting the latest EMC standard requirements.

FTM - Frankonia Turntable Series

Frankonia's wide range of turntables is fully compliant with the EMC chamber environment. The turntables are available in different sizes and can be equipped with various options. They are integrated flush in the raised floor and are surrounded by a conductivity grounding ring to ensure the contact with the ground plane of the chamber. For the control of the turntable series FTM, the Frankonia controller FC06.1 is perfectly adapted using IEEE 488.2 (GPIB) commands.

Range of Products

- › Integrated turntables FTM series from $\varnothing 1,2$ m to $\varnothing 12,0$ m and up to 80 tons
- › Integration of energy chains, rotary joints for data, electrics or fluids, exhaust and cooling systems, or customer specific items possible
- › Dynamometer, free-roller, or mobile on-top dynamometers can be integrated
- › FC06.1 controller with independent software (SCPI commands)
- › Controllable with common EMC software, and wireless access option



FAM - Frankonia Antenna Mast

The FAM is Frankonia's standard antenna mast solution compliant with CISPR 16-1-4. For customers' convenience, the Frankonia antenna mast is equipped with wheels for easy transportation and handling. To reduce unintentional reflections, the masts are made of fiberglass and plastic materials. Any reflecting materials have been reduced to a minimum.

FBM - Frankonia Boresight Mast

Based on the FAM, the FBM boresight antenna mast that includes an advanced tilt function is compliant with ANSI C63.4 and CISPR 16-1-4.

To offer full flexibility, the tilt function can be switched off, so that the FBM operates as a standard mast.

Within the tilt function, the FBM software automatically calculates the tilt angle in accordance with the antenna specific reference point, as well as distance, position and size of the EUT and monitors the complete test procedure.

Range of Products

- › FAM standard antenna mast (CISPR)
- › FBM boresight antenna mast (FCC/ANSI & CISPR)
- › FSM antenna stand, optional with polarization unit
- › FC06 controller with independent software (SCPI commands) incl. wireless access via smartphone or table
- › Controllable with common EMC software
- › Wireless interface to control from your handheld device

Advantages & Benefits

What makes our uniqueness and the perfect match

Frankonia is a top-notch solution provider for EMC and Antenna Test Solutions from Germany. We offer a well-developed and an all-encompassing range of distinctive products and unique absorbers to customize every project with perfection and time.

Product Advantages when selecting Frankonia

- › Modular and self-supporting system guarantees the highest shielding performance
- › Prefabricated standard paired with in-house production assures best quality
- › Extremely broad range of standard products and custom-specific products
- › In-house developed products are up to date considering latest technologies
- › Identical shielding performance throughout the whole system
- › Comprehensive automation devices from turntable to mast
- › Outstanding and unique Frankosorb® absorber technology as modular and interchangeable solution
- › Nothing is glued or welded; everything is mounted using bolting and screwing techniques
- › No timber works at all
- › No harmful materials at all (e.g., no carbon, no polyethylene, no glue)
- › Local adjustments to electricity, statics, technical parameters or building conditions are self-evident
- › An all-encompassing product line secures a single-point solution provider
- › All products are part of a well-developed system perfectly matched and comply with industrial standards

Customer Benefits when selecting Frankonia

- › Proven technologies and solutions since more than 35 years
- › Long-lasting and performance Frankosorb® absorbers
- › Safety to people, laboratory and building with A2 non-combustible absorbers
- › Just recyclable materials in use; everything produced in our modern and eco-friendly manufacturing
- › Overall, little building conditions and requirements to consider
- › Optimized design approaches for an easy and fast installation
- › Made-to-order arrangement of chamber with accessories that fits into your building
- › Simple interfaces to the building (electricity, ventilation, exhaust, data, gas, ...)
- › Improved handling of absorbers and antennas for faster throughput
- › Modular setup to re-located, modify or re-sell the complete chamber
- › Experienced project management cares from the tiny details to the big picture completing on point
- › Professional installation service and after-sales service by Frankonia

Why customers selecting Frankonia

- › Require perfection and reliability to all individual needs and for any project sizes
- › Consider future-proven and long-lasting products from a single source
- › Need building design support with cost-saving efforts
- › Expect latest technologies and materials, bearing in mind the safety for people and laboratory
- › Maximizing productivity and efficiency with optimal use of resources
- › Desire to get an outstanding lab at highest satisfaction

No compromises; just satisfaction throughout every custom-built setup. Frankonia provides authentic, proven and reliable solutions for the top-notch customer segment.

Frankonia Core Values

- › **Ingenuity**
We never stop pushing forward
We look for advantage in everything
We handle challenges with creativity
We strive for simplicity
- › **Energy**
We bring energy to everything we do
We stay hungry and believe in what we do
We are open to challenges and embrace opportunities
We share the passion for what we do
- › **Community**
We act fair and respectful
We roll up our sleeves to get the job done
We shape the future together
We win as a team
- › **Fairness**
We are open minded and curious
We admit failures, learn and improve
We communicate with clarity and honesty
We take responsibility for our actions

Sales & Information

sales@frankoniagroup.com
www.frankonia-solutions.com

Frankonia Germany EMC Solutions GmbH

Industriestr. 16, 91180 Heideck, Germany
sales@frankoniagroup.com Tel.: +49 (0) 9177 / 98 - 500

Frankonia EMC Test-Systems GmbH

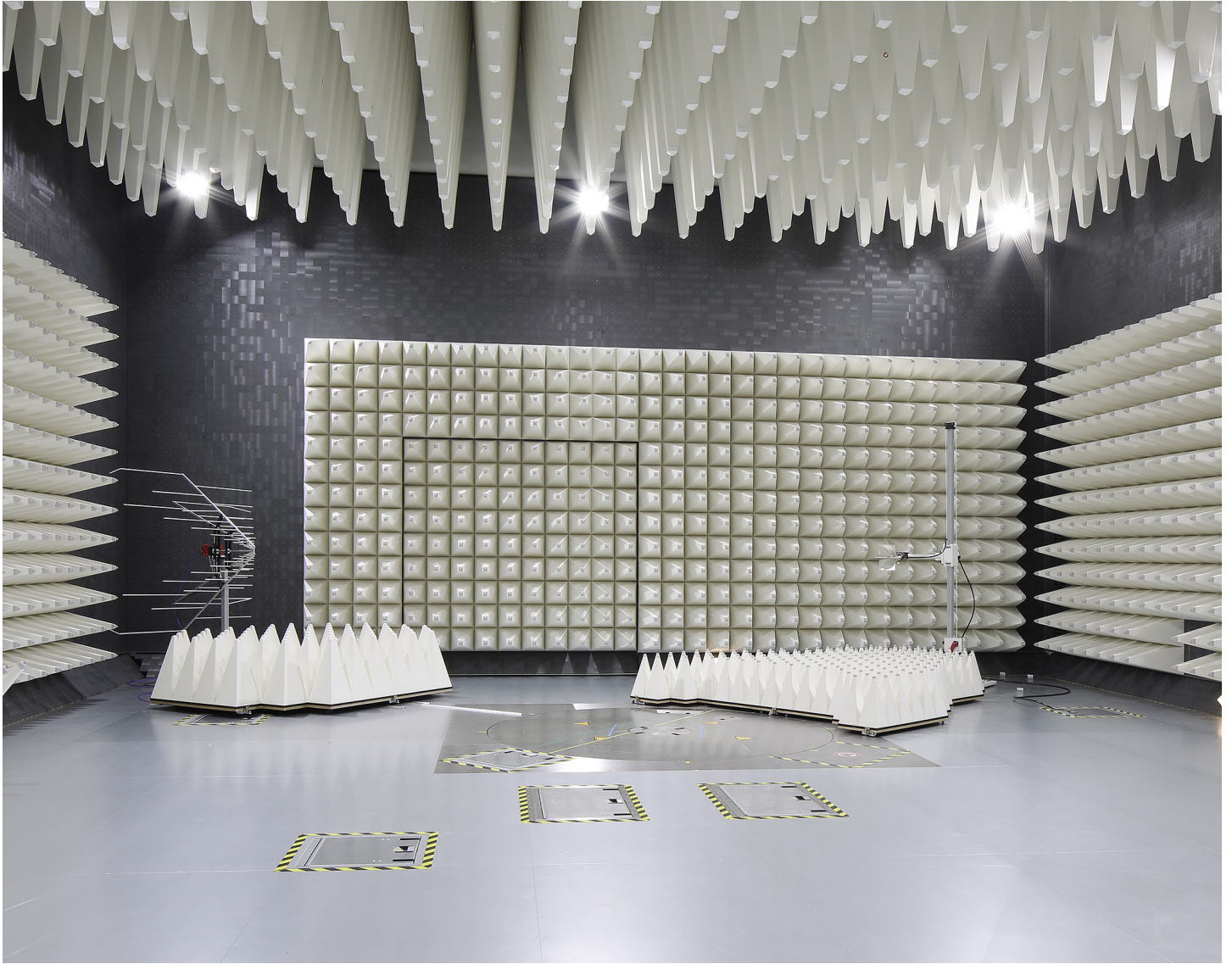
Daimlerstr. 17, 91301 Forchheim, Germany
sales@frankonia-emv.com Tel.: +49 (0) 9191 / 73 666 - 0

Jiashan Frankonia EMC Co., Ltd.

No.55, Hongqiao Rd, Zone 4, Jiashan Zhejiang 314100, China
info@emc-frankonia.com Tel.: +86 (0) 573 - 84731 555

Frankonia India EMC Solutions Pvt. Ltd.

18/1 Old 10/1 Lakshmanan Street, T. Nagar, Chennai 600017, India
sales@frankoniagroup.com Tel.: +91 (0) 44 28 153370



FRANKONIA

Your EMC Expert from Germany

Unique customer solutions for your EMC requirements



www.frankonia-solutions.com

sales@frankoniagroup.com